Exploring Excess Return Generation through European Fixed Income Arbitrages

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1 Introduction

Duarte et al. [2007] pose the question that motivates our analysis: do fixed-income arbitrage strategies earn genuine alpha, or are their returns simply compensation for systematic risk—"picking up nickels in front of a steamroller"? Using a ten-factor spanning regression, they find that the more complex strategies produce significant alpha that survives equity and bond market risk controls. We take this question to a new setting—European fixed-income markets—and to a broader set of strategies and factor controls than any prior study, namely the BTP Italia inflation-linked basis, the European corporate CDS–bond basis, and the iTraxx index skew (described in detail in Section 2). A second set of questions then arises: if the alpha is genuine, is it roughly constant, or is it time- or state-dependent? And, perhaps above all, why does it persist? The presidential address of Duffie [2010] provides a framework: when intermediary balance-sheet capacity is the binding constraint on arbitrage, mispricings across different markets should co-move, the co-movement should intensify in stress, and the transmission should follow a liquidity pecking order from more liquid to less liquid markets. Brunnermeier and Pedersen [2009] provide the mechanism—shocks to funding liquidity and market liquidity reinforce each other in a spiral that forces simultaneous deleveraging across positions—and Mitchell and Pulvino [2012] document the empirical counterpart, showing that arbitrage crashes propagate across strategies as capital withdraws.

Each of the three arbitrage strategies we study has an established literature, but none has been analysed in the way we propose. The inflation-linked bond basis was documented by Fleckenstein et al. [2014] in the context of US Inflation-Linked Treasury Bonds (TIPSs), who found an average mispricing of 54.5 basis points between TIPSs and nominal Treasuries over the 2004–2009 period, with peaks exceeding 200 basis points during the 2008–2009 financial crisis (Figure 8). This basis has since compressed substantially, oscillating in a 0–30 bps band post-2009(Figure 9), reflecting broader institutional participation in the U.S. inflation-linked market—and, as Fleckenstein et al. [2014] show, the basis narrows as hedge fund capital flows into the trade, providing direct evidence for the slow-moving capital channel. In their Section VII, they further document that changes in the

TIPS–Treasury mispricing are correlated with changes in the corporate CDS–bond basis and the CDX index–component basis, establishing the cross-strategy co-movement that we extend to the European setting in Section 5. In contrast, the analogous basis on Italian BTP Italia bonds persists to date. To the best of our knowledge, the BTP Italia has not been the subject of any prior academic study, and its inflation-linked basis has not previously been analysed as a systematic arbitrage strategy. Its persistence is particularly striking because the BTP Italia eliminates the ballooning risk that characterises TIPSs and other conventional inflation-linked sovereigns,[1] and because its retail-dominated investor base creates a structural segmentation that insulates the basis from the institutional capital flows that closed the U.S. mispricing.

Moving to the second strategy we analyse, the CDS–bond basis is the difference between the CDS spread and the bond credit spread for the same reference entity, a relationship that should be zero by no arbitrage in a frictionless market. It was first documented by Hull et al. [2004] and Longstaff et al. [2005] in its pre-crisis positive form. Mitchell and Pulvino [2012] show that during 2007–2009, tightening financial constraints drove the basis deeply negative, and Bai and Collin-Dufresne [2019] provide a comprehensive analysis of its cross-sectional determinants and its connection to limits of arbitrage. However, this literature focuses predominantly on U.S. corporate bond markets, and, to our knowledge, very little attention has been paid to the Euro counterpart, which is the second focus of this study.

This brings us to the third strategy of interest, namely the so-called CDS index skew, which is defined as the deviation of the traded index spread from the duration-weighted average of its single-name constituents. In a U.S. context, this strategy has been studied by Junge and Trolle [2015], who use it as a market-wide liquidity measure, and by Boyarchenko et al. [2016], who document that post-crisis regulatory capital requirements have made basis trades between the index and single-name CDS "significantly less economical," leading to persistently wider skews; both papers focus

[1]Conventional inflation-linked bonds adjust the principal upward with realised inflation; in deflationary scenarios or when inflation expectations spike, the redemption value can grow to many multiples of par (the so-called *ballooning* risk), introducing path-dependent valuation risk. The BTP Italia instead pays inflation revaluation as part of each semiannual coupon and redeems at par regardless of the inflation path, eliminating this source of uncertainty. The persistence of a positive arbitrage in the BTP Italia basis is therefore especially surprising: the instrument has *less* risk than a conventional inflation-linked bond, not more.

on the U.S. CDX index. To the best of our knowledge, neither the negative basis trade on European investment-grade corporates in the iTraxx Main universe nor the index skew across the four iTraxx sub-indices has been studied as a systematic arbitrage strategy in the academic literature.

Given this background, our paper extends the Duarte et al. [2007] framework in four respects:

1. Each of our three strategies has a deterministic payoff at maturity: if the position can be held to maturity, the profit is certain regardless of the path of market variables in the interim. The strategies in Duarte et al. [2007], by contrast—swap spreads, yield curve trades, mortgage hedging, volatility selling, capital structure trades—do not share this property: their payoffs depend on models, mean-reversion assumptions, or stochastic processes (prepayment, realised volatility) that leave the outcome uncertain even at maturity.

2. We subject the alpha estimates to a progressive battery of factor controls: three classical benchmark specifications from the fixed-income hedge fund literature, a rolling principal component analysis following Ludvigson and Ng [2009] on 85 candidate factors, and the Adaptive Elastic Net of Zou and Zhang [2009] with stability selection [Meinshausen and Bühlmann, 2010], which identifies a sparse set of risk drivers for each strategy individually. The alpha that survives this sequence of increasingly demanding tests cannot easily be attributed to omitted risk factors, and using the conditional framework of Ferson and Schadt [1996] we show that it rises systematically during periods of intermediary funding stress.

3. We test the cross-strategy implications of slow-moving capital systematically, drawing on Duffie [2010], Brunnermeier and Pedersen [2009], Mitchell and Pulvino [2012], and Fleckenstein et al. [2014]: whereas the existing literature has studied each of these arbitrages in isolation and predominantly in U.S. markets, we examine all three jointly in the European market and ask whether their mispricings co-move, whether the co-movement intensifies in stress, and whether a common factor tied to intermediary capital explains the co-movement. The European setting provides a natural experiment: two of our strategies (CDS–Bond and iTraxx skew) are traded by the same institutional intermediaries, while the third (BTP Italia) is held predominantly by retail investors, allowing us to test the slow-moving capital mechanism against a control group with a structurally different investor base.

4. We argue that post-crisis regulatory capital costs [Boyarchenko et al., 2016] have permanently raised the cost of intermediary balance-sheet capacity, and the heterogeneity in investor base across the three strategies generates distinct limits to arbitrage that the slow-moving capital literature [Duffie, 2010, Brunnermeier and Pedersen, 2009, Mitchell et al., 2007] predicts.

We find that all three strategies generate positive and economically significant excess returns net of transaction costs, with annualised Sharpe ratios from 0.71 (BTP Italia) to 1.71 (CDS–Bond Basis), positive skewness, and contained drawdowns. The alpha survives controls based on three widely used benchmark frameworks, on rolling principal components, and on the Adaptive Elastic Net for the BTP Italia and CDS–Bond strategies (the iTraxx skew, by contrast, is absorbed by its selected risk factors but serves as the leading indicator of intermediary funding stress in the cross-strategy analysis). The residual alpha rises during periods of intermediary funding stress, in a manner consistent with the slow-moving capital mechanism of Duffie [2010]. The three mispricings are linked by a common factor that loads on intermediary balance-sheet variables—the He et al. [2017] capital ratio, the Libor–OIS spread, and the Amihud and Mendelson [2015] illiquidity measure—and not on macroeconomic fundamentals (a placebo regression on macro indicators yields uniformly insignificant coefficients). The persistence of the mispricing levels mirrors the funding-dependence of each arbitrage trade: the CDS–Bond basis (which requires continuous repo financing of the cash bond) has a half-life of approximately four months, while the pure-derivative iTraxx skew converges in roughly one month. Within the institutional pair (CDS–Bond and iTraxx), correlations rise from $+0.10$ in low-stress to $+0.38$ in high-stress regimes; across investor bases, the BTP Italia correlation with the institutional strategies turns negative in stress, providing independent evidence of investor-base segmentation.

In light of the above, the paper is organized as follows. Section 2 describes the three strategies, their construction, and their performance. Section 3 evaluates the alpha against three classical benchmark frameworks. Section 4 extends the analysis to a candidate universe of 85 factors using

PCA and the Adaptive Elastic Net. Section 5 tests the cross-strategy predictions of the slow-moving capital literature. Section 6 concludes.

2 Data, Strategy Construction, and Performance

This section introduces the three European fixed-income arbitrage strategies studied in this paper: the inflation-linked basis on Italian government bonds (BTP Italia), the negative basis between corporate bonds and their credit default swaps (CDS–Bond Basis), and the CDS index skew—the difference between the traded iTraxx spread and the theoretical spread implied by the duration-weighted average of its single-name constituents (iTraxx Combined). Each strategy exploits a distinct relative-value mispricing in European markets, and each raises the question of whether the mispricing generates risk-adjusted returns that survive transaction costs and rigorous performance diagnostics.

We describe in Section 2.1 how each strategy is constructed, paying particular attention to the institutional features of the BTP Italia that distinguish it from conventional inflation-linked instruments; in Section 2.2 how individual trade returns are aggregated into investable indices, comparing equal-weighted and signal-weighted schemes and incorporating regime-dependent transaction costs; and in Section 2.3 the performance of the resulting strategy indices, verified against manipulation-proof and volatility-managed robustness checks.

2.1 Data and Strategy Construction

Data are sourced from the MOT market operated by Borsa Italiana (BTP Italia and nominal BTP prices), Bloomberg (inflation swap rates and corporate bond spreads), CMA / CME Group (single-name CDS spreads), and J.P. Morgan DataQuery (iTraxx index and constituent data, cross-validated against Barclays Live). The CDS–Bond and iTraxx skew samples begin in June 2004 with the launch of the first on-the-run series of the iTraxx Europe Main 5-year index. The BTP Italia sample begins with the first issuance in March 2012. All three samples extend to May 2025. Following Duarte et al. [2007], each individual trade is modelled as a separate fund that opens when the basis crosses an entry threshold and closes when the basis reverts through an exit threshold or the underlying instrument matures. Entry and exit rules are summarised in Table 1; trade-level statistics are reported in Table 2. Over the full sample, the strategies generate 260 trades for BTP Italia, 931 for CDS–Bond Basis, and 206 for iTraxx Combined. Figure 10 plots the market-level basis time series for each strategy.

BTP Italia: The Inflation-Linked Basis. The BTP Italia is an Italian government bond indexed to domestic inflation, measured by the ISTAT FOI consumer price index (excluding tobacco). It differs in two economically meaningful ways from conventional inflation-linked sovereigns such as the U.S. TIPS studied by Fleckenstein et al. [2014], and both differences are central to the arbitrage opportunity we document.

The first is the absence of *ballooning risk*. A conventional inflation-linked bond accumulates the inflation adjustment in the principal over the life of the bond, so that the notional outstanding at time t equals $100 \times I_t$, where I_t is the cumulative inflation factor since issuance. In a credit event, the recovery rate is applied to the original face value of 100, not to the inflation-adjusted notional: an investor holding a TIPS with $I_t = 1.25$ would recover $\mathrm{RR} \times 100$ and lose the entire accumulated revaluation of 25 percentage points. The BTP Italia eliminates this exposure entirely. The capital revaluation is paid out semiannually alongside the coupon, so the inflation factor resets to unity at every coupon date and the notional exposed to credit risk remains at par throughout the life of the bond. The inflation-linked leg of the arbitrage trade therefore becomes credit-equivalent to a nominal BTP, which isolates a pure relative-value component in the basis and removes the differential-default contamination that affects the TIPS–Treasury basis in stress periods. The technical specification of the indexation coefficient and the deflation floors on both coupon and principal is reported in Appendix A.1.

The second feature is the investor base. The Italian Treasury introduced the BTP Italia in 2012 explicitly as a retail instrument, structuring its primary placement in two sequential phases: an initial phase reserved exclusively for retail investors (which typically absorbs the bulk of each issuance), followed by a shorter institutional phase. Investors who purchase at issuance and hold to maturity receive an additional loyalty bonus (the *premio fedeltà*) that further incentivises buy-and-hold behaviour. The BTP Italia is therefore held predominantly by households with long holding periods, no balance-sheet constraints, and no mandate to arbitrage relative-value mispricings. This segmentation matters because, as we will document in Section 5, it is what makes the BTP Italia basis behave differently from the two institutional credit strategies during stress episodes: retail holders do not face margin calls and do not unwind their positions when intermediary capital withdraws, leaving the basis open to those arbitrageurs who can still deploy capital.

We construct the BTP Italia basis following the replication logic of Fleckenstein et al. [2014], adapted to the Italian instrument. The inflation-linked cash flows of the BTP Italia are converted into fixed-rate equivalents using the term structure of zero-coupon inflation swaps, yielding a fixed-rate equivalent yield y_t^{IL}. The basis is defined as the difference between this yield and the yield to maturity of the nominal BTP y_t^{NOM}:

$$\mathrm{Basis}_t^{\mathrm{BTP}} = y_t^{\mathrm{IL}} - y_t^{\mathrm{NOM}}. \tag{1}$$

A positive (negative) basis indicates that the BTP Italia, once converted to fixed, yields more (less) than the nominal bond at the same maturity. Operationally, the trade is executed through an asset swap (ASW) on the BTP Italia against an offsetting position in the maturity-matched nominal BTP: the strategy buys (sells) the asset-swapped BTP Italia and sells (buys) the nominal bond when the basis is positive (negative). Both the BTP Italia and the nominal BTP are accepted as general collateral in the euro repo market, so the two legs carry identical funding costs, which rules out funding-cost differentials as an explanation for the observed basis.

CDS–Bond Basis: The Negative Basis Trade. The CDS–Bond basis for issuer i at date t is defined as the difference between the CDS spread and the z-spread of a reference corporate bond:

$$\mathrm{Basis}_{i,t} = \text{CDS spread}_{i,t} - \text{z-spread}_{i,t}, \tag{2}$$

where the z-spread is computed as the constant spread that, when added to the bootstrapped euro zero-coupon swap curve, equates the discounted bond cash flows to the observed market price. When the basis is negative, credit protection via CDS is cheaper than the credit spread embedded in the bond price. The negative basis trade exploits this gap: the arbitrageur purchases the bond (earning the z-spread) and simultaneously buys CDS protection (paying the lower CDS premium), locking in a positive carry that persists until convergence or bond maturity. We restrict the strategy to the negative basis trade; the reverse trade (short bond, sell protection) requires sourcing the corporate bond through a reverse repo, which for corporate names is subject to recall risk—particularly during periods of stress, when the bond may be called back, forcing premature unwinding of the position [Bai and Collin-Dufresne, 2019]. In a default event, the bondholder recovers RR on par while the CDS contract pays $100 - \text{RR}$, so the combined position returns par regardless of realised recovery—provided the CDS maturity is no shorter than the bond's, which we ensure by matching hedge tenors to bond residual life (Appendix A.2). The tradable universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Europe Main 5-year index, which ensures liquidity and provides a transparent filter for investment-grade credit quality. When a name exits the on-the-run index at a roll date, no new trades are initiated on that name; existing positions are carried to convergence or maturity. All CDS are centrally cleared through LCH or ICE, eliminating direct counterparty credit risk on the protection leg.

iTraxx Combined: The CDS Index Skew. The CDS index skew is the difference between the traded spread of an iTraxx index and the theoretical spread implied by the duration-weighted average of its single-name constituents:

$$\text{Skew}_t = S_t^{\text{index}} - \sum_{i=1}^{N} w_{i,t} S_{i,t}^{\text{single}}, \tag{3}$$

where the duration weights $w_{i,t}$ ensure the replicating portfolio matches the aggregate credit risk sensitivity of the index. A positive (negative) skew indicates that the index trades wide (tight) relative to the sum of its parts. Three forces sustain a non-zero equilibrium skew. First, the index is far more liquid than its single-name constituents, especially in stress [Junge and Trolle, 2015]. Second, the two legs face different sources of demand: institutional investors use the index as the natural macro credit hedge, while dealer banks trade single names for more granular hedging needs, including the hedging of counterparty CVA exposure—forces that can push the two legs in non-synchronised directions and make the skew bidirectional. Third, executing the replicating basket of single-name positions carries operational costs that the index trade does not. Our strategy trades the skew across the four iTraxx sub-indices—Main, Senior Financials, Subordinated Financials, and Crossover—restricting new positions to the on-the-run series at the time of entry. Entry thresholds are calibrated to the historical volatility of each sub-index skew (Table 1). Unlike the CDS–Bond basis, the iTraxx skew is a derivative-on-derivative trade with no cash leg financed in repo. The two legs are traded under a single netting set—either centrally cleared at the same CCP or bilaterally with the same dealer under an ISDA Master Agreement and CSA—so that long and short positions offset for daily variation margin and balance-sheet purposes.

2.2 Index Construction and Weighting Schemes

Following Duarte et al. [2007], individual trades are treated as separate fictional hedge funds with their own entry date, exit date, and daily return stream, and the strategy index on any given day is the weighted average of the returns of the funds that are active on that day. We implement two weighting schemes. The equal-weighted (EW) scheme of Duarte et al. [2007] assigns identical weight $1/K_t$ to every active trade:

$$r_t^{\mathrm{EW}} = \frac{1}{K_t} \sum_{k=1}^{K_t} r_{k,t}. \tag{4}$$

The signal-weighted (SW) scheme, inspired by the curvature signal of Rebonato and Ronzani [2021], weights each trade proportionally to the absolute distance between the basis at entry and its historical equilibrium level,

$$r_t^{\mathrm{SW}} = \sum_{k=1}^{K_t} \frac{|b_{k,0} - \theta_{t_k}|}{\sum_{j=1}^{K_t} |b_{j,0} - \theta_{t_j}|} \, r_{k,t}, \tag{5}$$

where $b_{k,0}$ is the basis at the entry date t_k of trade k, θ_{t_k} is the expanding-window average of the cross-sectional basis from the start of the sample up to t_k, and $r_{k,t}$ is the signed return of trade k (positive when the position profits, with the direction—long or short—fixed at entry). The absolute value ensures that the weight reflects the magnitude of the dislocation rather than its sign, which is already carried by $r_{k,t}$. The rationale follows Rebonato and Ronzani [2021]: a basis that deviates further from its historical equilibrium represents a richer opportunity and warrants a larger allocation conditional on convergence. Signal-weighted indices dominate equal-weighted indices on both Sharpe and manipulation-proof performance for all three strategies (Appendix A.3, Table 4), and we adopt the SW scheme as the baseline throughout the paper; EW results in the appendix confirm that all findings are robust to this choice.

Market conditions are classified into three regimes based on the level of the iTraxx Europe Main 5-year spread: LOW (below 60 bps), MID (60–100 bps), and HIGH (above 100 bps). The same classification recurs in Section 3, where we decompose alpha by regime, and in Section 5, where we define stress episodes for the cross-strategy analysis. Transaction costs are applied at the individual trade level as entry and exit fees proportional to DV01, and fee levels are regime-dependent: bid-offer spreads and fees rise monotonically from LOW to HIGH, reflecting the empirical widening of dealer quotes during market dislocation. This tiered structure ensures that costs are conservative precisely in the stress periods when arbitrage returns are largest. The full fee schedule is reported in Table 3; from this point forward, all strategy returns are net of these regime-dependent costs.

2.3 Performance Analysis

Table 24 reports summary statistics computed from daily data. All strategy returns are excess returns by construction: the BTP Italia trade is self-financing (long and short legs are both repoed at the same general collateral rate), the CDS–Bond trade earns the z-spread net of the CDS premium (where the z-spread is itself measured over the swap curve, which proxies for the funding cost), and the iTraxx skew trade requires no capital beyond initial margin. Panel A describes the basis levels: all three series are highly persistent, with first-order autocorrelations between 0.87 and 0.98, consistent with the slow-moving capital literature [Duffie, 2010, Brunnermeier and Pedersen, 2009, Mitchell et al., 2007] that we examine formally in Section 5. Panel B describes the strategy returns. The CDS–Bond basis produces the highest annualised return (5.37%) and Sharpe ratio (1.71), with an annualised volatility of 3.15%. iTraxx Combined returns 1.96% p.a. with a Sharpe ratio of 0.76 over the longest sample of the three. BTP Italia earns 1.67% p.a. with a Sharpe ratio of 0.71 over a sample that begins with the first issuance in March 2012. All three strategies exhibit positive return skewness (0.60, 2.81, and 0.43 respectively), which cuts against the popular characterisation of fixed-income arbitrage as "picking up nickels in front of a steamroller" [Duarte et al., 2007]. Maximum drawdowns are moderate and short-lived, as Figure 11 illustrates: the deepest drawdown is 4.4% for the CDS–Bond basis during the 2008–2009 credit crisis, 3.0% for BTP Italia, and 2.5% for iTraxx Combined, and all three are recovered within months.

All three strategies generate positive and economically significant excess returns net of transaction costs, with annualised Sharpe ratios ranging from 0.71 (BTP Italia) to 1.71 (CDS–Bond Basis), positive skewness, and contained drawdowns. We subject this finding to two robustness checks below.

Manipulation-Proof Performance. Sharpe ratios are vulnerable to manipulation by strategies whose dynamic trading rules alter the shape of the return distribution, which is precisely the structure that defines the indices we construct. Table 25 therefore reports the Manipulation-Proof Performance Measure (MPPM) of Goetzmann et al. [2007] for risk-aversion parameters $\rho \in \{2,3,4\}$. The MPPM is the unique performance metric that satisfies four properties: it produces a single-valued score, is independent of dollar scale, cannot be enhanced by information-free trading or leverage, and is consistent with standard financial market equilibrium. These properties make it the appropriate benchmark for strategies, like ours, that use dynamic entry and exit rules.

The key finding is the remarkable stability of the MPPM across risk-aversion levels. For BTP Italia, the measure is 1.56, 1.55, and 1.54 percent per year at $\rho = 2,3,4$ respectively; for CDS–Bond Basis, 5.06, 5.01, and 4.97; for iTraxx Combined, 1.83, 1.82, and 1.80. The nearinvariance of the MPPM to ρ is a direct consequence of the low return variance and the limited drawdowns documented above. The MPPM penalises strategies that depend on extreme returns through the concave transformation $(1+r_t)^{1-\rho}$, so a strategy that earns its Sharpe ratio by occasionally suffering large losses sees its MPPM deteriorate sharply as ρ increases. Conversely, when returns are moderate and tails are thin, the transformation is approximately linear and the measure varies little with ρ. Read together with the positive return skewness documented above, the stability we observe indicates that the strategies' risk-adjusted performance does not rely on tail events: an investor with $\rho = 4$ evaluates these strategies almost identically to an investor with $\rho = 2$. This is strong evidence that the documented Sharpe ratios do not reflect compensation for unobserved tail risk that would dominate the assessment of a risk-averse investor.

Volatility-Managed Portfolios. A second concern that the literature has raised about fixed-income arbitrage is that unconditional Sharpe ratios may average over a time-varying risk-return trade-off, with periods of attractive compensation alternating with periods in which risk is poorly priced. Moreira and Muir [2017] propose a simple test of this hypothesis. The managed portfolio scales exposure inversely to the previous month's realised variance,

$$f_{t+1}^{\sigma} \;=\; \frac{c}{\hat{\sigma}_t^2} f_{t+1}, \tag{6}$$

where $\hat{\sigma}_t^2$ is the realised variance over month t and the scaling constant c is chosen so that the managed and unmanaged portfolios have the same unconditional standard deviation. A statistically significant managed-portfolio alpha relative to the buy-and-hold strategy is diagnostic of a time-varying ratio μ_t/σ_t^2: an investor can earn additional risk-adjusted returns by taking less risk precisely when volatility is high, which can only be rationalised if the risk-return trade-off is itself time-varying.

Table 26 reports results for the three strategies. The CDS–Bond basis produces a large and statistically significant managed-portfolio alpha of approximately 1.68% per year for the SW index, consistent with a time-varying funding-liquidity mechanism: when volatility is low, intermediary balance sheets are healthy and arbitrage capital is available to exploit the negative basis; when volatility spikes, capital withdraws and the trade-off deteriorates. For BTP Italia and iTraxx Combined, the managed-portfolio alphas are slightly negative (-0.22% and -0.16% p.a. respectively, both insignificant at any conventional level), indicating that the realised-variance signal does not identify exploitable variation in the risk-return trade-off for these two strategies. The contrast between strategies is itself informative and foreshadows the regime-dependent alpha analysis of Section 3, where we decompose alpha across stress regimes and show that the funding liquidity pattern documented here for the CDS–Bond basis generalises into a uniform regime effect across all three strategies once we control for benchmark factor exposures.

Having established that all three strategies generate positive and robust risk-adjusted returns net of transaction costs, we turn next to a formal assessment of these alpha estimates against the canonical benchmark factor models for fixed-income arbitrage.

3 Benchmark Factor Models

This section estimates three canonical benchmark spanning regressions from the fixed-income arbitrage literature—Duarte et al. [2007], Fung and Hsieh [2004], and Brooks et al. [2020]—and tests whether the alpha of the three strategies survives once their systematic factor exposures are controlled for. A central question, motivated by the slow-moving capital framework of Duffie [2010], is whether the residual alpha varies systematically with intermediary funding stress. We describe in Section 3.1 the three specifications and the construction of their euro-denominated counterparts, and present in Section 3.2 the regression results, the cross-model alpha synthesis, the regime decomposition, and the conditional alpha model of Ferson and Schadt [1996].

3.1 Benchmark Specifications

Three benchmark specifications are widely used in the empirical literature on fixed-income arbitrage and active fixed-income management, and they provide the natural reference points against which our strategies should be evaluated.

Duarte et al. [2007], in the paper that motivates our research design, test whether fixed-income arbitrage excess returns are compensation for exposures to equity, banking-sector, and term-structure risk:

$$\begin{aligned} r_{i,t} \;=\; & \alpha_i + \beta_1\, Mkt_t + \beta_2\, SMB_t + \beta_3\, HML_t + \beta_4\, UMD_t + \beta_5\, RS_t + \beta_6\, RI_t \\ & + \beta_7\, RB_t + \beta_8\, R2_t + \beta_9\, R5_t + \beta_{10}\, R10_t + \varepsilon_{i,t}, \end{aligned} \tag{7}$$

where *Mkt*, *SMB*, *HML*, and *UMD* are the Fama–French–Carhart equity factors (market, size, value, and momentum), *RS* is the S&P bank stock index excess return, *RI* and *RB* are A/BAA-rated industrial and bank bond index excess returns, and *R2*, *R5*, *R10* are the CRSP Fama bond portfolios at two-, five-, and ten-year maturities.

Fung and Hsieh [2004] propose a seven-factor *asset-based style* model that tests whether hedge fund returns are explained by equity, fixed-income, and trend-following exposures:

$$\begin{aligned} r_{i,t} \;=\; & \alpha_i + \beta_1\, S\&P_t + \beta_2\, SC\text{-}LC_t + \beta_3\, BdOpt_t + \beta_4\, FXOpt_t \\ & + \beta_5\, ComOpt_t + \beta_6\, \Delta 10Y_t + \beta_7\, \Delta CredSpr_t + \varepsilon_{i,t}, \end{aligned} \tag{8}$$

where *S&P* is the S&P 500 excess return, *SC-LC* is the small-minus-large-cap total return, *BdOpt*, *FXOpt*, and *ComOpt* are the three trend-following factors constructed as lookback straddle returns on bond, currency, and commodity futures, Δ*10Y* is the change in the 10-year Treasury yield, and Δ*CredSpr* is the change in the Baa–Treasury credit spread.

Brooks et al. [2020] take a different angle, decomposing the returns of active fixed-income managers into eight *traditional risk premia* organised along three channels of risk-taking—duration, credit, and volatility selling:

$$\begin{aligned} r_{i,t} \;=\; & \alpha_i + \beta_1\, Term_t + \beta_2\, GlTerm_t + \beta_3\, GlAgg_t + \beta_4\, InflLink_t + \beta_5\, CorpCred_t \\ & + \beta_6\, EmDebt_t + \beta_7\, EmCurr_t + \beta_8\, USTVol_t + \varepsilon_{i,t}, \end{aligned} \tag{9}$$

where *Term* and *GlTerm* are the excess returns on US and global Treasury total return indices, *GlAgg* is the excess return on a global aggregate bond index, *InflLink* is the excess return on a global inflation-linked bond index, *CorpCred* is a 50/50 blend of a US high-yield corporate bond index (in excess of duration-matched Treasuries) and a leveraged loan index (in excess of 3-month LIBOR), *EmDebt* is the duration-adjusted excess return on an emerging-market debt index, *EmCurr* is the return on an equal-weighted basket of emerging-market currencies, and *USTVol* is the return on delta-hedged straddles on 10-year Treasury futures.

Each benchmark is estimated in two parallel panels. The first uses the original U.S.-denominated factors as specified by the respective authors, which is the natural way to replicate each framework as it was first published. The second uses euro equivalents, since all three strategies trade European instruments and may therefore be better explained by European factor exposures.

For the euro panel, the Fama–French factors—used directly in Duarte et al. [2007] and indirectly in Fung and Hsieh [2004]—are sourced from the Kenneth French data library; because these are reported in USD, we convert them to EUR following Glück et al. [2021]. For long factors (market excess return):

$$F_t^{\mathrm{EUR}} = \frac{1 + F_t^{\mathrm{USD}} + r_{f,t}^{\mathrm{USD}}}{1 + r_t^{\mathrm{USD/EUR}}} - 1 - r_{f,t}^{\mathrm{EUR}}, \tag{10}$$

and for long-short factors (SMB, HML, UMD):

$$LS_t^{\mathrm{EUR}} = \frac{LS_t^{\mathrm{USD}}}{1 + r_t^{\mathrm{USD/EUR}}}, \tag{11}$$

where F_t^{USD} and LS_t^{USD} are the original factors in US dollars, $r_{f,t}^{\mathrm{USD}}$ and $r_{f,t}^{\mathrm{EUR}}$ are the US and euro risk-free rates (Fama–French T-bill rate and one-month Euribor, respectively), and $r_t^{\mathrm{USD/EUR}}$ is the period return on the dollar-to-euro exchange rate. All remaining factors enter the euro panel with native European counterparts: European bank stocks, euro-area corporate bond indices, and German government bonds in the Duarte specification; euro-area term, credit, and inflation-linked premia in Brooks et al.; and the 10-year Bund yield and Baa–Bund credit spread in Fung and Hsieh. The Fung and Hsieh trend-following lookback straddle factors are currency-independent by construction (D. Hsieh, private communication) and enter both panels identically. The complete factor list is reported in Appendix A.4.1.

All specifications are estimated by ordinary least squares with Newey–West HAC standard errors [Newey and West, 1987], with the truncation lag set as $\lfloor T^{1/4} \rfloor$ following Newey and West [1994]. Variance inflation factors for each specification are reported in Appendix A.9. All factors exhibit VIF below 10 except for the Fama bond portfolios (R2, R5, R10) in the Duarte specification (VIF $\approx$ 15–55), which reflects the well-known correlation structure of maturity-sorted Treasury returns.

3.2 Results and Discussion

Factor Model Regressions. Tables 27, 28, and 29 report the full regression results. Across all three frameworks, all three strategies, and both currency denominations, the estimated intercept is positive and statistically significant at the one-percent level. The annualised alphas are economically meaningful: approximately $+5\%$ p.a. for the CDS–Bond basis across all specifications, $+1.6$–3.5% for iTraxx Combined, and $+1.7$–2.1% for BTP Italia, consistent with the unconditional Sharpe ratios documented in Section 2.3.

The adjusted R^2 is low across all specifications, ranging from one to eighteen percent, indicating that the standard factor sets leave most of the return variation unexplained—a question we take up in Section 4. The significant factor loadings that do emerge differ across strategies and are economically intuitive. The CDS–Bond basis loads on credit-related factors: bank bond returns and the Baa–Treasury credit spread change are significant across the Duarte and Fung–Hsieh specifications. BTP Italia and iTraxx Combined load instead on equity factors—market, size, and value in Duarte, S&P 500 and small-minus-large cap in Fung–Hsieh—consistent with the indirect equity market exposure that these strategies carry. The Brooks et al. specification, which uses broader risk premia rather than individual asset returns, produces fewer significant loadings but delivers comparable alpha estimates. Despite these significant loadings, the low $\bar{R}^2$ indicates that the bulk of excess return variation remains unexplained, motivating the data-driven factor selection of Section 4.

Cross-Model Comparison and Regime Analysis. Table 30 consolidates the alpha estimates from the three frameworks and decomposes them by market regime. Panel A reports the unconditional alphas side by side: for each strategy, all three frameworks deliver positive and significant alpha, and the estimates are consistent across specifications despite their different factor structures.

Panel B decomposes alpha into two sub-samples defined by the level of the iTraxx Europe Main 5-year spread, using the same regime classification introduced in Section 2.3: HIGH-stress months are those in which the spread exceeds 100 basis points, and NORMAL the remaining observations. In every cell of the table, alpha during HIGH-stress months is substantially larger than alpha during NORMAL months, typically by a factor of two to four. The pattern holds across all three benchmark specifications and across all three strategies, and it does not depend on the precise threshold used to define the HIGH regime: in Appendix A.7 we report the same exercise at threshold levels of 80, 100, and 120 basis points, with similar results.

This regime dependence is consistent with the theoretical mechanism that Duffie [2010] formalises in his presidential address. When intermediary capital is abundant, arbitrage forces are strong and mispricings are quickly closed, leaving little room for excess returns beyond what compensates for the systematic factor loadings. When intermediary capital is scarce—as it is during episodes of widening credit spreads—the arbitrageurs who ordinarily close the gap retreat from the trade or are forced to unwind existing positions, the mispricings widen, and the residual premium for those who can still deploy capital rises sharply. Our finding that the alpha component unexplained by benchmark factors is precisely the component that grows in stress is consistent with the slow-moving capital mechanism. It is also the empirical pattern that motivates the cross-strategy analysis in Section 5, where we test whether the same intermediary balance-sheet shocks drive co-movement of mispricings across the three strategies.

Conditional Alpha. Panel C of Table 30 formalises the regime effect using the conditional performance evaluation framework of Ferson and Schadt [1996], which lets the alpha intercept vary continuously with a publicly observable state variable rather than across a binary regime split. The specification is

$$r_{i,t} \;=\; \alpha_0 \;+\; \alpha_1 z_t \;+\; \beta' \mathbf{X}_t \;+\; \varepsilon_t, \tag{12}$$

where $r_{i,t}$ is the strategy excess return, $\mathbf{X}_t$ collects the factor returns of the chosen benchmark, and z_t is the standardised level of the iTraxx Europe Main 5-year spread: $z_t = (s_t - \bar{s})/\hat{\sigma}_s$. The state variable z_t is constructed to be mean-zero and unit-variance over the sample, so the intercept α_0 retains its interpretation as the unconditional alpha and the slope α_1 measures the additional alpha earned per one-standard-deviation increase in funding stress. Standard errors are again computed with the Newey and West [1994] HAC correction. The Ferson and Schadt [1996] formulation has two advantages over the binary split in Panel B. It uses the full continuous variation in the stress proxy rather than collapsing it into a dummy, which is more efficient when the underlying state variable is meaningfully time-varying; and it makes the intermediary funding mechanism explicit by tying the movement in alpha to a single, economically interpretable proxy for dealer balance-sheet health.

The estimates of α_1 in Panel C are positive for all three strategies and all three benchmark specifications, and the unconditional intercept α_0 remains significant in every case, confirming that a baseline alpha persists even after conditioning on the stress state. For CDS–Bond Basis and iTraxx Combined, α_1 is significant at the one-percent level across all three frameworks: a one-standard-deviation increase in the iTraxx Main spread raises CDS–Bond alpha by approximately three percentage points per year and iTraxx alpha by one to three percentage points. For BTP Italia, α_1 is positive in all specifications but weaker in magnitude, consistent with the shorter sample available for this strategy. For all three strategies, the implied alpha at $z = +1\sigma$ is two to four times larger than at $z = -1\sigma$, indicating that the bulk of the unconditional alpha is earned during periods of elevated funding stress. Reading Panel B and Panel C together, the unconditional alpha masks a pronounced time variation consistent with the slow-moving capital mechanism of Duffie [2010].

Rolling Alpha. A natural concern with regime-based or conditional analyses is that the estimated relationship between alpha and stress may be driven by a single sub-period—perhaps the global financial crisis of 2008–2009, or the European sovereign debt crisis of 2011–2012, or the COVID-19 shock of March 2020—rather than by a stable structural feature of the data. Figure 12 addresses this concern directly by plotting rolling 36-month alpha estimates for each strategy under each of the three benchmark specifications. The visual evidence is clear: in every panel, the rolling alpha is positive almost everywhere, peaks visibly during the shaded HIGH-stress episodes, and returns to a lower but still positive level during normal periods. The spikes are not localised in any single sub-period; they recur whenever the iTraxx Main spread crosses the HIGH threshold. The three lines within each panel—one for each benchmark—track each other closely, which reinforces the cross-model robustness already documented in Panel A of Table 30. Detailed sub-period splits and threshold robustness for each framework individually are reported in Appendix A.7.

Taken together, the four sets of results in this section establish two findings that are central to the rest of the paper. First, the alpha of the three arbitrage strategies is robust: it survives controls based on three widely used benchmark frameworks for fixed-income arbitrage, in both currency denominations, and it does so for every cross-section and every sub-period that we examine. Second, the alpha is not constant. It rises systematically when intermediary capital is scarce, consistent with the slow-moving capital theory, and the magnitude of this regime effect is large enough to dominate the unconditional Sharpe ratio in several specifications. The first finding indicates that the observed excess returns are not explained by the systematic exposures that the prior literature has identified. The second finding points to a deeper economic mechanism, but it does not yet identify which specific risk factors are responsible for the residual variation. We turn to that question in Section 4, where we use penalised regression methods on a much larger candidate factor universe to identify the factors that these three benchmarks omit.

4 Factor Selection and Spanning Regressions

The benchmark factor models of Section 3 are designed to capture known systematic exposures, but they leave open the question of whether a broader set of risk factors can better explain strategy excess returns and absorb the residual alpha. Starting from a candidate universe of 85 risk factors assembled in Section 4.1, we proceed in two steps. We begin in Section 4.2 with a rolling principal component analysis following Ludvigson and Ng [2009], which extracts a small number of latent factors from this candidate set and tests whether these diffuse combinations span strategy excess returns. We then turn in Section 4.3 to the Adaptive Elastic Net of Zou and Zhang [2009], a penalised regression that selects individual observable factors from the same candidate set, and we compare its output with alternative machine learning methods in Section 4.5. As in Section 3, we examine whether the alpha that survives factor controls varies with intermediary funding stress.

4.1 Candidate Factor Universe

We assemble a candidate set of 85 risk factors from a comprehensive review of the empirical asset pricing, intermediary, and macro-finance literatures. Each factor is selected on the basis of a published theoretical or empirical link to fixed-income arbitrage returns, intermediary balance-sheet constraints, or broad market risk conditions. The factors are organised into six economic categories: credit risk (Panel A), liquidity and funding (Panel B), equity (Panel C), volatility (Panel D), macro-financial (Panel E), and interest rates and term structure (Panel F). The complete list with definitions, data sources, and references is reported in Table 5 (Appendix A.4.1). This candidate universe serves as the common input to both the PCA of Section 4.2 and the penalised regressions of Section 4.3.

4.2 Principal Component Analysis

When the number of candidate factors is large relative to the sample size—as is the case with 85 variables and approximately 160–240 monthly observations—direct regression is infeasible and the standard approach in the empirical macro-finance literature is to extract a small number of principal components that summarise the common variation in the factor set [Ludvigson and Ng, 2009]. We implement a rolling PCA that estimates factor loadings on a trailing window of $W = 24$ months, standardises each factor within the estimation window, and projects the data at the evaluation date onto the estimated loadings to obtain out-of-sample principal component scores. This rolling design avoids look-ahead bias: the loadings used to construct the scores at time t are estimated entirely from data available before t. We retain $K = 8$ components, which capture approximately eighty percent of the cumulative variance in the candidate factor matrix (Figure 13). Robustness of the results to alternative (W, K) combinations is reported in Appendix A.5.1.

The spanning regression takes the form

$$r_{i,t} \;=\; \alpha_i \;+\; \beta_i' \mathbf{PC}_t \;+\; \varepsilon_{i,t},$$

where $r_{i,t}$ is the monthly excess return of strategy i and $\mathbf{PC}_t$ is the $K \times 1$ vector of rolling principal component scores at time t. We estimate the regression using contemporaneous timing ($\mathbf{PC}_t \to r_{i,t}$) as the baseline, with predictive timing ($\mathbf{PC}_t \to r_{i,t+1}$) as a robustness check; both timings yield similar conclusions. Inference uses Newey–West HAC standard errors.

Table 31 reports the results under both contemporaneous and predictive timing. Alpha remains statistically significant at the one-percent level for all three strategies under both conventions: $+1.61\%$ p.a. for BTP Italia, $+5.87\%$ for CDS–Bond, and $+2.21\%$ for iTraxx Combined under contemporaneous timing, with similar magnitudes under predictive timing. The adjusted $\bar{R}^2$ is low across all specifications, reaching at most 12% under contemporaneous timing and falling below zero under predictive timing for BTP Italia, indicating that eight diffuse principal components do not span strategy excess returns. Table 32 reports the average loadings of the retained components: PC1 is dominated by credit and equity market factors, PC2 by global rates and bond returns, and PC3 by swap spreads and dealer CDS. Because PCA is an unsupervised method—it extracts factors that maximise variance in the candidate set, not factors that explain variation in strategy returns—

it cannot identify strategy-specific risk drivers. This motivates the use of a supervised penalised regression in the next subsection.

4.3 Adaptive Elastic Net

Preprocessing. The 85 candidate factors described in Section 4.1 are preprocessed for penalised regression. Stationarity is verified via augmented Dickey–Fuller tests, and non-stationary series are first-differenced. Following the normalisation convention of Zou and Zhang [2009], each factor is standardised to unit L_2-norm rather than unit variance, which ensures that the penalisation treats all factors symmetrically regardless of scale. Near-duplicate pairs with absolute correlation exceeding 0.95 are identified and the member of each pair with the lower marginal correlation to the strategy excess return is dropped, reducing the effective dimension to approximately 82 factors per strategy. Even after this reduction, the space of possible factor subsets—$2^{82} \approx 4.8 \times 10^{24}$—rules out exhaustive search and motivates the use of penalised estimation methods with formal variable selection guarantees.

Estimation. The Adaptive Elastic Net of Zou and Zhang [2009] proceeds in two stages. In Stage 1, a standard Elastic Net is estimated by solving

$$\hat{\beta}^{\text{EN}} = \arg\min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^{T} \left(r_t - \alpha - \beta' \mathbf{f}_t \right)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \|\beta\|_1 \right\}, \tag{13}$$

which combines an ℓ_2 ridge penalty that handles collinearity with an ℓ_1 LASSO penalty that induces sparsity. The Elastic Net produces a preliminary estimate $\hat{\beta}^{\text{EN}}$ that is used to construct adaptive weights $w_j = |\hat{\beta}_j^{\text{EN}}|^{-\gamma}$ with $\gamma = 1$. Factors with large Stage 1 coefficients receive small weights (weak penalisation), while factors with near-zero Stage 1 coefficients receive large weights (strong penalisation), so the adaptive weighting concentrates the penalty on the irrelevant factors and relaxes it on the relevant ones. In Stage 2, the Adaptive Elastic Net re-solves with factor-specific ℓ_1

penalties:

$$\hat{\beta}^{\text{AEN}} = \arg\min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^{T} \left(r_t - \alpha - \beta' \mathbf{f}_t \right)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^{p} w_j |\beta_j| \right\}. \tag{14}$$

Both stages use the same λ_2; the pair (λ_1, λ_2) is selected by minimising the Hannan–Quinn information criterion (HQC), and the solution path is computed via coordinate descent [Friedman et al., 2010].

The AEN is preferred to the standard LASSO for two reasons that are directly relevant to our setting. First, the LASSO does not possess the oracle property: Zou and Zhang [2009] show that under conditions on the design matrix that are routinely violated when candidate factors are correlated—precisely the situation in our data—the LASSO can be inconsistent for variable selection. The adaptive weighting restores consistency and delivers the oracle property, meaning that the AEN performs asymptotically as if the true set of relevant factors were known in advance. Second, the ℓ_2 ridge component handles the grouping effect: when several candidate factors are highly correlated (as are, for example, the various credit spread measures in our universe), the LASSO arbitrarily selects one and drops the rest, while the Elastic Net tends to select or drop them together, producing more stable and economically interpretable factor sets.

Stability Selection. Even with the oracle property, the factors selected by a single run of the AEN on a finite sample may be sensitive to small perturbations in the data. To address this selection uncertainty, we apply the stability selection framework of Meinshausen and Bühlmann [2010], adapted for time-series dependence. The procedure draws $B = 100$ stationary bootstrap resamples with an expected block length of 9 months, re-estimates the full AEN pipeline on each resample, and records which factors are selected. A factor is retained in the final "stable" set only if its selection frequency across the 100 bootstrap replicates exceeds $\hat{\pi}_j \geq 80\%$. Factors that are selected occasionally but not consistently are discarded as unreliable.

The key theoretical result of Meinshausen and Bühlmann [2010] is that stability selection controls the expected number of false selections: for a threshold $\pi^* \in (1/2, 1)$ and a maximum model size q per bootstrap replicate, the expected number of falsely selected variables is bounded by $q^2/[p\,(2\pi^* - 1)]$, where p is the total number of candidates. With $p \approx 82$ factors and a threshold of 80%, this bound is small, providing a formal guarantee against overfitting that no single-run LASSO or Elastic Net can offer. Because correlated factors may split bootstrap votes among themselves (each individually falling below the threshold even though the cluster as a whole is consistently selected), we additionally report cluster-level selection frequencies in Appendix A.6.2.

Post-Selection OLS. Once the stable factor set $\hat{S}_i$ has been identified for each strategy, we discard the penalised coefficient estimates and re-estimate the spanning regression by unrestricted OLS on the original (un-standardised) factors:

$$r_{i,t} \;=\; \alpha_i \;+\; \beta_i' \mathbf{f}_{\hat{S}_i,t} \;+\; \varepsilon_{i,t}, \tag{15}$$

with Newey–West HAC standard errors. This two-step procedure—penalised selection followed by unpenalised inference—avoids the well-known downward bias of penalised coefficient estimates while retaining the sparsity and interpretability of the selected model. The post-selection OLS regression is the basis for all alpha estimates, factor loadings, and $\bar{R}^2$ values reported in Table 33.

Results. Table 33 reports the post-selection OLS estimates. Alpha is statistically significant at the one-percent level for BTP Italia (+1.39% p.a.) and CDS–Bond Basis (+3.94% p.a.); for iTraxx Combined, the alpha is positive but not statistically significant (+0.47%, $t = 1.04$), suggesting that the skew strategy's excess returns are largely explained by the selected factors. The models are parsimonious—six, eight, and five factors respectively—yet deliver adjusted $\bar{R}^2$ of 21.7%, 26.7%, and 23.6%, substantially higher than either the benchmark frameworks of Section 3 or the PCA spanning regressions above.

The selected factors are economically interpretable and differ across strategies in ways that reflect their distinct market microstructures. For BTP Italia, the stable set includes the Amihud and Mendelson [2015] illiquidity measure, the Libor–OIS spread, the 10-year swap spread, the CDX

Investment Grade index, and two equity style factors (SMB, momentum)—a combination that points to liquidity, funding conditions, and interest rate exposures as the primary systematic drivers of the inflation-linked basis trade. For CDS–Bond, the selected factors include financial uncertainty (ΔUF, Ludvigson et al., 2021), European investment-grade bond returns (RI_EU), emerging-market currency risk, the 1-year European prime broker CDS spreads (PB_EU_CDS_1Y), the 5-year swap spread, U.S. credit spread changes, the option-implied equity premium (EP_SVIX, Martin, 2017), and the currency trend-following factor—consistent with a trade whose profitability depends on dealer balance-sheet capacity, funding conditions, and cross-market risk appetite. For iTraxx Combined, the dominant factors are the option-implied equity premium (EP_SVIX), the Libor–OIS spread, European equity volatility (ΔV2X), real uncertainty (ΔUR, Ludvigson et al., 2021), and U.S. Treasury settlement fails—reflecting the index skew's sensitivity to the aggregate state of intermediary capital and market-wide uncertainty rather than to idiosyncratic credit events. Each strategy thus loads on a distinct combination of risk drivers, consistent with the view that the alpha documented in Sections 3 and 4 is not compensation for a single common factor but reflects strategy-specific exposures that only a targeted selection method can identify. To verify that the results do not depend on the choice of penalised estimator, we re-estimate the spanning regression using four alternative methods—LASSO, standard Elastic Net, Adaptive LASSO, and the ALASSO-Chen variant of Chen et al. [2025]—and report the comparison in Section 4.5. The full list of factors selected by each method is reported in Appendix A.6.2.

4.4 Conditional Alpha

Table 34 decomposes alpha by stress regime under both the PCA and AEN factor specifications. The CDS–Bond basis exhibits the sharpest contrast: under AEN factors, alpha rises from $+4.15\%$ (NORMAL) to $+8.17\%$ (HIGH), nearly doubling while remaining significant at the one-percent level in both regimes; the PCA specification delivers a similar pattern ($+4.84\%$ to $+9.13\%$). For BTP Italia, the HIGH-regime alpha is substantially larger under both PCA ($+5.37\%$, significant at the five-percent level) and AEN ($+2.71\%$), but the AEN estimate does not reach statistical significance ($t = 1.43$), reflecting the smaller number of HIGH-stress months in the shorter BTP Italia sample. For iTraxx Combined, the alpha is significant in both regimes under PCA factors (+3.38% in HIGH, +1.82% in NORMAL), while under AEN factors it is significant only during HIGH stress (+1.51%) and turns negative and insignificant during NORMAL months—indicating that the AEN factors absorb the skew's unconditional return but not the stress-related component. This regime dependence is consistent with the slow-moving capital framework of Duffie [2010]: when intermediary balance sheets are under pressure, mispricings widen and the residual excess return rises. Detailed Ferson–Schadt conditional alpha estimates and threshold robustness checks (at 80, 100, and 120 bps) are reported in Appendix A.5.3 and A.6.3. Sub-period stability of the alpha estimates under both PCA and AEN is confirmed in Appendix A.5.2, and rolling 36-month alpha estimates with regime shading are plotted in Appendix A.6.4.

4.5 Discussion, Method Comparison, and Robustness

Table 35 compares alpha and model dimension across five machine learning methods: LASSO, Elastic Net, Adaptive LASSO, AEN (our baseline), and ALASSO-Chen [Chen et al., 2025]. The first four methods use HQC for hyperparameter selection; ALASSO-Chen uses AIC, which imposes a lighter penalty and therefore selects substantially more factors. For BTP Italia, the LASSO, Elastic Net, and Adaptive LASSO converge to the same seven-factor model under HQC tuning; the AEN stability selection drops one factor, yielding a six-factor specification with nearly identical alpha (+1.39% p.a., significant at the one-percent level). For CDS–Bond Basis, alpha ranges from +3.91% to +4.97% across methods, always significant at the one-percent level. For iTraxx Combined, the alpha is not statistically significant under any method ($t = 1.04$–1.68), confirming that the skew strategy's excess returns are absorbed by the selected risk factors—a finding that is itself informative, as it indicates that the iTraxx skew does not offer a genuine risk-adjusted premium beyond its systematic exposures. The ALASSO-Chen specification selects 51 factors for BTP Italia and raises $\bar{R}^2$ to 48%, but the alpha loses statistical significance ($t = 1.62$), suggesting overfitting: the additional factors absorb noise rather than genuine risk exposures. AEN is preferred because it combines the oracle property of Zou and Zhang [2009] with the stability selection filter of Meinshausen and Bühlmann [2010], producing the sparsest model that retains only factors with robust bootstrap selection frequencies. Sensitivity of the factor selection to the correlation pre-cleaning threshold (ρ) and the adaptive weight exponent (γ) is verified in Appendix A.6.1: in all configurations, the AEN selects the same core factors with identical post-selection OLS results.

Having established that alpha survives both diffuse (PCA) and targeted (AEN) factor controls for the BTP Italia and CDS–Bond strategies, that the iTraxx skew is absorbed by its selected risk factors, and that the residual alpha rises during stress in a manner consistent with Duffie [2010], we turn next to the question of whether the three mispricings are linked to one another and, if so, what economic mechanism connects them.

5 Cross-Strategy Interdependencies

The preceding sections have established that all three strategies generate alpha that survives both classical benchmark and data-driven factor controls, and that this alpha rises during periods of intermediary funding stress. A natural question follows: are the three mispricings linked to one another, and if so, what economic mechanism ties them together? The slow-moving capital literature provides the theoretical framework from which we derive five testable predictions. The qualitative anchor is Duffie [2010], who argues that mispricings serviced by a common pool of intermediary capital co-move and that distortions are amplified when balance-sheet capacity is constrained. Brunnermeier and Pedersen [2009] formalize this intuition in a model with funding-constrained speculators, deriving Proposition 6 on the commonality of market liquidity, the simultaneity of liquidity jumps ("commonality of fragility"), and the asymmetric impact of capital shocks across securities of different liquidity. Mitchell and Pulvino [2012] document the empirical mechanism, showing that forced deleveraging propagates across arbitrage strategies on a horizon of several months. Finally, Fleckenstein, Longstaff, and Lustig [2014] provide the methodological template for testing cross-arbitrage co-movement through spanning regressions on contemporaneous and lagged mispricing changes.

From this combined framework we derive five testable predictions: (P1) mispricings serviced by a common pool of intermediary capital should co-move; (P2) the co-movement should intensify when that capital is scarce; (P3) mispricing changes in one strategy should contain information for predicting those of others, with effects propagating asymmetrically along the liquidity gradient; (P4) the persistence of mispricing levels should reflect the funding-dependence of each arbitrage trade, with strategies that require continuous balance-sheet funding displaying the longest half-lives [Boyarchenko et al., 2016]; and (P5) a common factor extracted from the co-movement should be explained by intermediary balance-sheet variables rather than by macroeconomic fundamentals. This section tests each of these predictions.

We present in Section 5.1 the correlation structure of the three mispricings across stress regimes (P1, P2). We then estimate in Section 5.2 spanning and interaction regressions (P2, P3), AR(1) persistence of the mispricing levels (P4), and a regression of the common factor on intermediary funding proxies (P5). Section 5.3 identifies Granger-causal transmission and traces the impulse response to shocks through the system, providing complementary evidence on P3. Section 5.4 interprets the evidence and explains why these arbitrage opportunities persist in post-crisis European credit markets.

5.1 Correlations, Co-Widening, and Persistence

Mispricing Construction. Following Duffie [2010], we work with mispricing *magnitudes* $m_{i,t}$ rather than strategy returns, because it is the level of the mispricing—the distance of the basis from its no-arbitrage value—that measures the degree to which capital is failing to close the gap. For each strategy, $m_{i,t}$ is defined as the absolute value of the basis, sign-adjusted so that an increase $\Delta m > 0$ denotes a widening of the mispricing for all three strategies. The BTP Italia mispricing is the cross-sectional mean of the absolute basis across all outstanding issues with at least six months to maturity; the CDS–Bond mispricing is the mean of the twenty most negative single-name bases in the iTraxx Main universe on each day; and the iTraxx Combined mispricing is the mean of the absolute skew across the four on-the-run sub-indices. All three series are aggregated to monthly frequency. Market conditions are classified into three regimes—LOW, MEDIUM, and HIGH—based on the level of the iTraxx Europe Main 5-year spread, with thresholds at 60 and 120 basis points.

Unconditional and Regime-Dependent Correlations. Table 36 reports pairwise Pearson correlations for both strategy returns (Panel A) and mispricing changes Δm (Panel B), computed unconditionally and separately within each of the three stress regimes. The first prediction of Duffie [2010]—that mispricings serviced by a common pool of intermediary capital should co-move—finds support in the corporate credit cluster: the correlation between CDS–Bond Δm and iTraxx Δm is $+0.18$ unconditionally and rises from $+0.10$ in the LOW regime to $+0.38$ in the HIGH regime, consistent with the theory. When dealer balance sheets are healthy (LOW regime), the correlation is moderate; when they are depleted (HIGH regime), the correlation becomes strongly positive as common funding shocks force the simultaneous unwinding of positions in both markets. The second feature of Table 36 is the behaviour of the BTP Italia pairs. The correlation between BTP Italia and iTraxx Δm moves from $+0.09$ in LOW to -0.53 in HIGH, reversing sign as stress increases; the CDS–Bond vs BTP Italia correlation similarly deepens from -0.17 to -0.35. This two-cluster pattern—positive and rising within the institutional cluster, negative and falling for the retail–institutional pairs—we interpret in Section 5.4. Figure 14 visualises the regime contrast for all three pairs. Robustness of these findings to purging by common risk factors, to Forbes–Rigobon bias correction, and to DCC-GARCH time-varying correlations is confirmed in Appendix A.8.

5.2 Spanning Regressions Across Strategies

Cross-Strategy Spanning. We test whether mispricing changes in one strategy predict those of another by estimating spanning regressions following the design of Fleckenstein et al. [2014]:

$$\Delta m_{i,t} \;=\; \alpha + \sum_{j \neq i} \sum_{l=0}^{L} \beta_{j,l} \, \Delta m_{j,t-l} + \varepsilon_t, \tag{16}$$

where $\Delta m_{i,t}$ is the monthly change in mispricing for strategy i and $L = 2$ lags. Inference uses Newey–West HAC standard errors.

Table 37 reports the results. The $\bar{R}^2$ is 9.7% for CDS–Bond, 13.3% for BTP Italia, and 5.5% for iTraxx Combined. Significant lagged coefficients reveal lead-lag transmission across strategies, consistent with a liquidity pecking order: the iTraxx index, the most liquid instrument in the system, adjusts first; the effect then propagates to the less liquid CDS–Bond cash market and finally to the BTP Italia inflation-linked basis, with a delay of one to two months. BTP Italia is the most predictable ($\bar{R}^2 = 13.3\%$), indicating that the corporate credit cluster has meaningful predictive power for the inflation-linked basis despite the two-cluster segmentation documented above. This pattern is consistent with the funding-liquidity feedback of Brunnermeier and Pedersen [2009], in which a shock to speculator capital propagates more strongly to illiquid securities, and with the cross-strategy capital reallocation documented empirically by Mitchell and Pulvino [2012]: when dealer capital is withdrawn from the credit markets, the ripple effect reaches the inflation-linked market with a delay that reflects the time it takes for the capital constraint to propagate across the dealer network.

Stress Amplification. The second prediction of Duffie [2010] is that cross-strategy transmission should intensify when intermediary capital is under pressure. The model is explicit on this point: when balance-sheet capacity is depleted, intermediaries are "less able or predisposed to absorb supply and demand shocks," and the resulting price distortions are larger and more correlated across markets. We test this prediction with an interaction regression:

$$\Delta m_{i,t} \;=\; \beta_0 \Delta m_{j,t} \;+\; \beta_1 \left(\Delta m_{j,t} \times z_t\right) \;+\; \gamma z_t \;+\; \varepsilon_t, \tag{17}$$

where z_t is the standardised iTraxx Main 5-year spread, which proxies for intermediary funding stress. The coefficient β_1 captures the *incremental* transmission during stress: $\beta_1 > 0$ means that a given shock to Δm_j has a larger effect on Δm_i when intermediary balance sheets are strained.

Table 38 reports the results. Within the corporate credit cluster, the interaction coefficients carry the predicted positive sign and the direct stress effect is significant for the CDS–Bond basis at the five-percent and one-percent levels respectively. For the pairs involving BTP Italia, however, the interaction coefficient is significantly *negative* ($\beta_1 = -1.27$ for iTraxx $\rightarrow$ BTP Italia, significant at the five-percent level): stress does not amplify the transmission from the institutional strategies to BTP Italia but rather *widens* the gap between the two clusters. This asymmetry is consistent with the Duffie framework when the investor bases of the two markets are structurally different. The institutional strategies are serviced by leveraged intermediaries whose balance sheets co-move with the stress proxy; the BTP Italia is held by retail investors who are not subject to the same funding pressures and therefore do not participate in the correlated unwinding.

Persistence and Funding-Dependence. The fourth prediction tests whether the persistence of mispricings is linked to the funding-dependence of each arbitrage trade. Following Boyarchenko et al. [2016], who argue that the post-crisis equilibrium level of basis trades reflects the regulatory cost of intermediary balance-sheet capacity, we expect the most persistent mispricings to be those whose execution requires continuous funding of a cash position on dealer balance sheets. We estimate first-order autocorrelations of the mispricing levels $m_{i,t}$ via

$$m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}, \qquad i \in \{\text{CDS–Bond}, \text{BTP Italia}, \text{iTraxx}\}, \tag{18}$$

with Newey–West standard errors (six monthly lags). The implied half-life is $h_i^* = \ln(0.5)/\ln(\phi_i)$.

Table 39 reports the estimates. The CDS–Bond basis is the most persistent series ($\phi = 0.85$, half-life ≈ 4.1 months), the BTP Italia is intermediate ($\phi = 0.68$, half-life ≈ 1.8 months), and the iTraxx skew is the least persistent ($\phi = 0.57$, half-life ≈ 1.2 months). All three coefficients are highly significant under HAC inference ($t > 6$). This ordering is consistent with the funding-dependence interpretation: the CDS–Bond basis requires the arbitrageur to hold the cash bond on a repo book and finance it continuously, so any widening persists until dealers can deploy fresh balance-sheet capacity [Boyarchenko et al., 2016, Mitchell and Pulvino, 2012]; the iTraxx index skew, by contrast, is a pure derivative basis with no continuous funding requirement, and converges within one to two months once a discrepancy is observed across the dealer network. The intermediate persistence of the BTP Italia reflects the regular refresh of the basis through periodic primary auctions reserved to households, which limits the persistence of any individual mispricing event.

Common Factor Structure. Duffie [2010] argues that the common factor driving correlated mispricings across markets should reflect the state of intermediary balance sheets—"the level of capital commonly available to bear losses"—rather than macroeconomic fundamentals. Brunnermeier and Pedersen [2009] formalise this claim in their Proposition 6, which states that the commonality of market liquidity across assets is driven by the speculators' shadow cost of capital. Table 40 tests this prediction by regressing the first principal component (PC1) of the three Δm series on two complementary sets of variables: (i) intermediary stress proxies organised by the three channels through which slow-moving capital operates, and (ii) a placebo set of macroeconomic indicators that should *not* explain PC1 if the slow-moving capital interpretation is correct.

Table 40 reports the results. Panel A selects one proxy per channel from the intermediary asset pricing literature: the He et al. [2017] capital ratio (HKM_IC: -4.29, significant at the five-percent level) as a measure of aggregate intermediary capital, the Libor–OIS spread ($+3.34$, significant at the one-percent level) as a measure of global interbank funding cost, European prime broker CDS spreads as a direct gauge of dealer credit risk (-0.018, significant at the five-percent level), and the Amihud and Mendelson [2015] illiquidity measure as a proxy for the cost of executing the arbitrage trade ($+1.73$, significant at the one-percent level). All four carry the sign predicted by the theory: a decline in intermediary capital, a rise in funding costs, and an increase in market illiquidity all predict wider mispricings. The model $\bar{R}^2$ is 23.1%. Panel B confirms these findings with an extended set of proxies that includes the GC-repo–T-bill spread, the European Bloomberg Financial Conditions Index, and settlement fails. Panel C reports the placebo regression of PC1 on macroeconomic indicators (macro and real-activity uncertainty, long-term inflation expectations, and economic policy uncertainty): *none* of the four coefficients is statistically significant (p-values ranging from 0.19 to 0.98), the model $\bar{R}^2$ falls to 11.1%, and the model itself is not jointly significant (F-statistic p-value $= 0.23$). Together, the three panels provide direct evidence that the common factor driving the co-movement of the three mispricings is a balance-sheet variable of the intermediary sector rather than a macroeconomic fundamental, consistent with the slow-moving capital theory of Duffie [2010] and the funding-liquidity model of Brunnermeier and Pedersen [2009].

5.3 VAR Analysis and Impulse Responses

Granger Causality. The spanning regressions of Section 5.2 document that mispricing changes in one strategy predict those of another, but they do not identify the direction of the transmission. We address this with Granger causality tests embedded in a VAR on the three Δm series. The results reveal a clear transmission hierarchy: the iTraxx index—the most liquid instrument in the system—Granger-causes BTP Italia at the one-percent significance level, and the CDS–Bond basis also Granger-causes BTP Italia at the five-percent level. Neither BTP Italia nor CDS–Bond Granger-causes iTraxx in the reverse direction with comparable strength. This pecking order is consistent with a world in which funding shocks originate in the standardised, exchange-traded credit index market (where dealer inventory adjustment is fastest), then propagate to the less liquid OTC cash-CDS basis, and finally reach the retail-dominated inflation-linked market where the response is delayed by the slower adjustment speed of a segmented investor base. Robustness to lag length ($p = 1, 2, 3$) is confirmed in Appendix A.8.3.

Generalized Impulse Responses. Figure 15 plots Generalized Impulse Response Functions [Pesaran and Shin, 1998], which have the advantage of being invariant to the ordering of variables in the VAR and therefore avoid the identification assumptions required by the Cholesky decomposition. The cross-responses provide complementary evidence on the asymmetric transmission documented in P3. A one-standard-deviation shock to iTraxx Δm generates a significant *negative* response in BTP Italia Δm that persists for approximately two to three months before dissipating, while the reverse direction (BTP Italia $\rightarrow$ iTraxx) is statistically indistinguishable from zero. The negative sign of the cross-response is consistent with the mechanism documented by Mitchell and Pulvino [2012]: when forced to delever, intermediaries sell their more liquid positions first to generate cash, leaving the less liquid retail-held BTP Italia basis unaffected by the institutional unwinding. The own-responses (diagonal panels) all decay rapidly within the first month at monthly frequency, which is the natural counterpart of the much higher persistence documented for the levels of the mispricings in P4 (Table 39): innovations to Δm dissipate quickly because the level m is mean-reverting on a multi-month horizon, with the speed of mean-reversion governed by the funding-dependence of each arbitrage trade. Robustness of the Granger causality results to lag length ($p = 1, 2, 3$) and to alternative stress proxies (iTraxx Crossover, V2X, VIX) is confirmed in Appendix A.8.

5.4 Discussion: Evidence for Slow-Moving Capital

Investor-Base Segmentation as a Natural Experiment. Although all three arbitrage strategies are implemented by the same type of institutional intermediary, the underlying instruments differ sharply in who holds them. The CDS–Bond basis and the iTraxx skew are arbitrage trades on instruments that are themselves held and traded by other institutional participants—hedge funds, dealer proprietary desks, and leveraged accounts—so that both the arbitrageur and the natural counterparty are subject to common funding constraints. When a spike in volatility inflates the Value-at-Risk of the combined position, the risk budget of the trading desk is exhausted, and forced liquidation occurs across both the cash-CDS and index-skew books simultaneously [Brunnermeier and Pedersen, 2009]. The resulting withdrawal of risk capacity pushes both the negative basis and the index skew wider at the same time. The BTP Italia, in contrast, is a sovereign instrument whose primary placement and secondary holdings are structurally dominated by retail investors. The Italian Treasury introduced the bond in 2012 with a primary auction reserved to households and a loyalty bonus (*premio fedeltà*) for buy-and-hold investors, and the security itself serves as the simplest available hedge against domestic inflation for an Italian household. The result is a bondholder base that behaves as a preferred-habitat investor in the sense of Vayanos and Vila [2021]: retail holders do not use leverage, do not face margin calls, and do not unwind their positions when dealer capital is scarce. While the arbitrageur of the BTP Italia basis is institutional and faces the same funding-liquidity environment as the arbitrageur of the CDS–Bond basis, the absence of forced selling on the bondholder side prevents the BTP Italia basis from participating in the institutional fire sales that widen the corporate credit pair during stress, and explains the negative correlation with CDS–Bond and iTraxx that intensifies from LOW to HIGH stress regimes (Table 36, Figure 14). This bondholder-base segmentation constitutes a natural experiment that tests the slow-moving capital mechanism: if the co-movement of the three mispricings were driven by a common macroeconomic factor, all three would move in the same direction regardless of who holds the underlying bond.

Why Do These Arbitrages Persist? A natural question for any paper that documents persistent arbitrage returns is: why have these opportunities not been competed away? The evidence in this section suggests that the answer lies in a combination of structural limits to arbitrage that differ across strategies but share a common root in the scarcity of intermediary capital.

For the BTP Italia, the binding constraint is investor-base segmentation. The retail holders who dominate the primary market neither monitor nor exploit the relative-value mispricing between the inflation-linked and the nominal bond. Institutional arbitrageurs can and do exploit the basis, but the retail holders who constitute the natural counterparty neither monitor the relative-value mispricing nor have the tools to arbitrage it, so the mispricing persists [Vayanos and Vila, 2021].

For the CDS–Bond basis, the limits are those studied by Bai and Collin-Dufresne [2019] and Mitchell and Pulvino [2012]: the trade requires funding the corporate bond on dealer balance sheets, the repo market for European investment-grade corporates is thin and subject to recall risk, and the counterparty credit risk on the CDS leg consumes regulatory capital. The result is a mechanism of procyclical capacity and countercyclical mispricing: when the economy is strong, dealer balance sheets are healthy and the basis is tight; when the economy deteriorates, balance-sheet capacity contracts, arbitrage capital withdraws, and the basis widens—precisely when the trade is most attractive but least feasible to execute.

For the iTraxx skew, the constraint is post-crisis regulatory capital. Boyarchenko et al. [2016] document that the supplementary leverage ratio (SLR) and other Basel III requirements have increased the capital charge for basis trades between the index and its single-name constituents, reducing the return on equity from approximately 6–9% pre-crisis to 2–3% post-crisis. Previously attractive trades have become, in the words of Boyarchenko et al. [2016], "significantly less economical," and the resulting withdrawal of dealer capital has led to "potentially 'new normal' levels for both the CDS-bond and the CDX-CDS bases." This regulatory channel is particularly important for our paper because it implies that the mispricings we document are not transient anomalies that will be closed by the natural forces of competition, but structural features of a market in which the cost of intermediary capital has been permanently raised by regulation. The persistence of alpha documented in Sections 4 and 5 is therefore not a puzzle that calls for a risk-based explanation; it is the equilibrium outcome of a market in which slow-moving capital, in the precise sense of Duffie [2010], keeps mispricings open longer than frictionless models would predict [Mitchell et al., 2007].

Mapping the Evidence to the Five Predictions. We close with an explicit mapping of our empirical findings to the five testable predictions that emerge from the slow-moving capital literature [Duffie, 2010, Brunnermeier and Pedersen, 2009, Mitchell and Pulvino, 2012, Boyarchenko et al., 2016, Fleckenstein et al., 2014].

P1. *Cross-market correlation.* Mispricings serviced by a common pool of intermediary capital should co-move. The unconditional correlation between CDS–Bond and iTraxx mispricing changes is $+0.18$ (Table 36): when one basis widens, the other tends to widen as well, consistent with both strategies drawing on the same pool of dealer capital.

P2. *Stress amplification.* The co-movement should intensify when intermediary capital is scarce. The pairwise correlation within the institutional cluster rises from $+0.10$ in the LOW regime to $+0.38$ in the HIGH regime (Table 36), and the interaction regressions show that the direct effect of stress on CDS–Bond mispricing is significant at the five-percent level or better (Table 38): a common funding shock widens both bases simultaneously, and the effect is stronger when balance sheets are already strained.

P3. *Lead-lag transmission.* Mispricing changes in one strategy should contain information for predicting those of others, with effects propagating asymmetrically along the liquidity gradient. The spanning regressions (Table 37), following the design of Fleckenstein et al. [2014], show that lagged iTraxx mispricing changes predict BTP Italia mispricing changes one to two months ahead, and Granger causality tests confirm that iTraxx leads BTP Italia at the one-percent level. This pattern is consistent with the funding-liquidity transmission of Brunnermeier and Pedersen [2009] and the cross-strategy capital reallocation documented by Mitchell and Pulvino [2012]: capital arrives first in the most liquid, standardised index market and reaches the less liquid cash and inflation-linked markets with a delay reflecting the time required for the capital constraint to propagate across the dealer network.

P4. *Persistence reflects funding-dependence.* The persistence of mispricing levels should reflect the funding-dependence of each arbitrage trade. AR(1) estimation on the monthly mispricing series (Table 39) yields a half-life of approximately four months for the CDS–Bond basis (the only strategy that requires continuous repo financing of the cash bond), approximately two months for the retail-held BTP Italia, and roughly one month for the iTraxx index skew (a pure derivative basis with no funding requirement). All three autocorrelation coefficients are highly significant under HAC inference. This ordering, consistent with Boyarchenko et al. [2016] and Mitchell and Pulvino [2012], links persistence directly to the regulatory cost of dealer balance-sheet capacity.

P5. *Common factor = intermediary capital, not macro fundamentals.* The factor driving cross-strategy co-movement should be explained by intermediary balance-sheet variables, not by macroeconomic fundamentals. PC1 of the three mispricing series is regressed on intermediary stress proxies (Table 40, Panel A): the He et al. [2017] capital ratio, the Libor–OIS spread, the European prime broker CDS spread, and the Amihud and Mendelson [2015] illiquidity measure are all significant with the predicted sign, and together they explain 23.1% of the variation in the common factor. A placebo regression on macroeconomic indicators (Panel C) yields four uniformly insignificant coefficients ($p > 0.19$) and a model that is not jointly significant, confirming that the common factor is a balance-sheet variable of the intermediary sector rather than a real-activity or price-level fundamental.

All five predictions find support in the data. The geography of slow-moving capital, it turns out, is determined not by asset-class labels but by the network of intermediaries that service these markets—and ultimately by who holds the bonds.

Having documented both the existence of significant alpha across three distinct European fixed-income arbitrage strategies and the economic mechanism—slow-moving intermediary capital, amplified by post-crisis regulation and modulated by investor-base segmentation—that sustains these mispricings, we draw our conclusions in Section 6.

6 Conclusion

We document positive and economically significant alpha in three European fixed-income arbitrage strategies—the BTP Italia inflation-linked basis, the corporate CDS–bond basis, and the iTraxx index skew—and show that this alpha cannot be attributed to known systematic risk factors. The mispricings are linked by a common factor that loads on intermediary balance-sheet variables (and not on macroeconomic fundamentals), and their persistence at the level of each series reflects the funding-dependence of the arbitrage trade: the CDS–Bond basis, the only strategy requiring continuous repo financing of the cash bond, is the most persistent, while the pure-derivative iTraxx skew is the least. The three correlations exhibit an asymmetric pattern that mirrors the structural divide between institutional and retail investor bases: positive and rising in stress within the institutional pair (CDS–Bond and iTraxx), and negative across investor bases (BTP Italia vs. either institutional strategy). The slow-moving capital literature [Duffie, 2010, Brunnermeier and Pedersen, 2009, Mitchell and Pulvino, 2012], combined with the post-crisis regulatory environment documented by Boyarchenko et al. [2016], provides an economic explanation for why these mispricings persist.

The analysis has three main limitations. First, the BTP Italia sample begins in March 2012 and is therefore short by the standards of the asset pricing literature, which constrains the power of regime-dependent tests for this strategy. Second, our identification of investor-base segmentation rests on the documented retail dominance of the BTP Italia primary market; we do not observe the cross-sectional flow of retail orders and therefore cannot directly test the preferred-habitat channel suggested by Vayanos and Vila [2021]. Third, our cross-strategy VAR is estimated on the post-2012 common sample, which excludes the 2008–2009 financial crisis but covers the post-Basel III regime in which the regulatory capital constraints of Boyarchenko et al. [2016] are binding—arguably the more relevant environment for the "new normal" of intermediary-driven mispricings, but a restriction worth noting.

The findings have implications for two strands of the literature. For the limits-to-arbitrage literature, the European setting provides a natural test of the slow-moving capital mechanism in a market with a structurally heterogeneous investor base, and the two-cluster correlation pattern offers independent evidence that the geography of arbitrage capital—rather than asset-class labels—determines cross-market co-movement. For the fixed-income asset pricing literature, the persistence of alpha after Adaptive Elastic Net controls on a large candidate factor universe suggests that the factor zoo, while comprehensive, may miss strategy-specific risk drivers that only supervised selection methods can identify.

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A Appendix

A.1 BTP Italia: Indexation Coefficient and Deflation Floors

The BTP Italia indexes both coupons and principal to the Italian consumer price index for blue- and white-collar worker households, excluding tobacco (FOI ex tabacchi) published by Istat. At each coupon date t, the indexation coefficient CI_t is the ratio of the relevant FOI reference index to its value at the previous coupon date $t-6\mathrm{m}$, so that CI_t measures the cumulative inflation over the six-month coupon period. The semiannual coupon and the capital revaluation payment are computed as

$$C_t = \tfrac{1}{2} c N \widetilde{\mathrm{CI}}_t, \qquad R_t = N\left(\widetilde{\mathrm{CI}}_t - 1\right), \tag{19}$$

where c is the fixed real annual coupon rate set at issuance, N is the subscribed nominal capital, and $\widetilde{\mathrm{CI}}_t$ is the *modified* indexation coefficient that incorporates a semiannual deflation floor:

$$\widetilde{\mathrm{CI}}_t = \max\bigl(\mathrm{CI}_t, 1\bigr). \tag{20}$$

If the FOI ex tabacchi falls between consecutive coupon dates, the indexation coefficient is reset to one for that semester, the coupon reverts to the fixed real rate $c/2 \cdot N$, and no capital revaluation is paid; in the following semester, the base index is reset to the higher of the previous semester's level and the highest reading observed in earlier semesters. The principal is redeemed at par at maturity (N), regardless of the cumulative inflation path. Two features of this design are relevant for the arbitrage. First, the semiannual reset of $\widetilde{\mathrm{CI}}_t$ to unity at every coupon date means that the notional exposed to default risk is always the original face value N, eliminating the path-dependent contamination that affects conventional inflation-linked bonds such as the U.S. TIPS or the euro-area BTP€i. Second, the deflation floor on the coupon guarantees a strictly positive nominal payment in every period, which is absent from instruments with floor only at maturity.

A.2 Trade Construction Details

A.2.1 Entry/Exit Rules and Thresholds

Table 1 summarises entry and exit thresholds for each strategy. All strategies require a minimum of 6 months to maturity at entry and maintain at least one open trade after the first signal. The CDS–Bond universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Main series; names that exit the index are no longer traded. All CDS are centrally cleared. The iTraxx skew strategy trades only on-the-run series across Main, SnrFin, SubFin, and Xover.

Table 1
Entry and Exit Thresholds by Strategy

Strategy	Entry Long	Entry Short	Exit Long	Exit Short
BTP Italia	> 40	< -50	< 10	> -10
CDS–Bond Basis	< -40	–	> 0	–
iTraxx Main	> 5	< -10	< -5	> 10
iTraxx SnrFin	> 8	< -15	< -5	> 5
iTraxx SubFin	> 10	< -20	< -10	> 10
iTraxx Xover	> 15	< -25	< -15	> 10

All thresholds in bps. CDS–Bond Basis is a one-directional negative basis trade (long bond, buy CDS protection). Minimum 6 months to maturity at entry for all strategies. Trades shorter than 3 trading days excluded ex-post.

A.2.2 Trade-Level Statistics

Table 2 reports trade-level statistics for each strategy, including the number of trades, average duration, average profit per trade, and the distribution of exit reasons.

A.2.3 Transaction Costs

Transaction costs are computed at the individual trade level and applied as entry and exit fees proportional to DV01 (bonds) or duration (CDS). Fee levels are regime-dependent: at each trade date, the prevailing level of the iTraxx Main 5Y index determines which of three tiers applies (Table 3). Thresholds are set at 60 and 100 bps, reflecting the empirical widening of bid-ask spreads in

Table 2
Trade-Level Statistics by Strategy

	BTP-Italia	iTraxx *combined*	CDS-Bond Basis
Total trades	260	206	931
Long	224	146	931
Short	36	60	0
Avg duration (days)	386	225	240
Median duration (days)	212	97	119
Avg P&L per trade (%)	2.079	1.491	2.463
Median P&L per trade (%)	1.960	1.395	1.927
Exit reasons (%)			
Target hit	92.3	97.1	90.7
Maturity	6.9	0.5	1.0
End of sample	0.8	2.4	8.4

Note: This table reports trade-level statistics for each arbitrage strategy. P&L is the cumulative return per trade in percentage points. Duration is in calendar days. Exit reasons may not sum to 100% due to rounding.

stressed markets. For trades held to maturity, the exit fee is waived. All return series in subsequent sections are net of these costs.

Table 3
Transaction Fee Schedule (bps per unit DV01 or duration)

	LOW (Main $<$ 60)	MID (60–100)	HIGH (Main $>$ 100)
BTP Italia	3	5	8
CDS–Bond Basis	4	7	10
iTraxx Main	0	1	1
iTraxx SnrFin	0	1	1
iTraxx SubFin	1	2	3
iTraxx Xover	3	5	7

Fee applied as: entry DV01 $\times$ fee_bps at entry $+$ exit DV01 $\times$ fee_bps at exit. Exit fee waived for trades held to maturity. Regime determined by iTraxx Main 5Y level at trade date: LOW ($<$60 bps), MID (60–100 bps), HIGH ($>$100 bps).

A.3 Signal-Weighted vs Equal-Weighted Comparison

Table 4 compares the signal-weighted (SW, Rebonato and Ronzani, 2021) and equal-weighted (EW, Duarte et al., 2007) indices. SW is adopted as the baseline in the main text; EW results confirm robustness to the weighting scheme.

Table 4
Summary Performance Metrics: Equal-Weighted vs Signal-Weighted

Strategy	Return (% p.a.)	Vol (% p.a.)	Sharpe	MaxDD (%)	Skew	Kurt	$\Theta_{\rho=3}$ (% p.a.)	N
Panel A: EW								
BTP-Italia	1.41	2.49	0.57	-2.99	0.54	19.76	1.32	159
CDS-Bond Basis	4.55	2.89	1.57	-4.47	2.54	53.35	4.42	203
iTraxx *combined*	1.37	2.05	0.67	-1.63	0.40	7.19	1.31	243
Panel B: SW								
BTP-Italia	1.67	2.36	0.71	-2.99	0.60	22.85	1.59	159
CDS-Bond Basis	5.37	3.15	1.71	-4.43	2.81	46.91	5.23	203
iTraxx *combined*	1.92	2.59	0.74	-2.52	0.43	5.66	1.82	243

Note: Metrics computed from daily returns (annualization factor 252). MaxDD is computed on the wealth index. $\Theta_{\rho=3}$ is the Goetzmann et al. (2007) manipulation-proof measure.

A.4 Candidate Factor Universe

A.4.1 Complete Factor List with Sources

Table 5 reports the full set of candidate risk factors used in the penalised regressions Section 4, together with their data sources, transformations, and economic category.

Table 5
Candidate Risk Factors

Factor ID	**Description**	**Reference**
	Panel A: Credit Risk Factors	
BTP_BUND	Yield differential between the Italian 10-year BTP and the German Bund.	Fausch and Sigonius [2018]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 5 (continued)

Factor ID	Description	Reference
CDX_IG	First difference of the CDS spread on the CDX Investment Grade index.	Klaus and Rzepkowski [2009]
CORP_CREDIT	Broad corporate credit factor. Excess return on high-yield and leveraged loan indices over duration-matched Treasuries.	Brooks et al. [2020]
CREDIT_EU	Bond credit risk factor. Yield spread between euro-area BBB and AAA corporate bond indices.	Fama and French [1989]
CREDIT_US	Bond credit risk factor. Yield spread between Moody's BAA and AAA corporate bond indices.	Fama and French [1989]
CRED_SPR_EU	Change in the credit spread between euro-area BAA corporate bond yields and 10-year German Bund yields.	Fung and Hsieh [2001]
CRED_SPR_US	Change in the credit spread between Moody's BAA yields and 10-year U.S. Treasury yields.	Fung and Hsieh [2001]
DEF_EU	Bond default risk factor. Total return spread between long-term European corporate bonds and long-term German Bunds.	Fama and French [1993]
DEF_US	Bond default risk factor. Total return spread between long-term U.S. corporate bonds and long-term U.S. government bonds.	Fama and French [1993]
EBP	Excess Bond Premium. Residual component of aggregate corporate bond spreads after netting out expected default risk and bond characteristics via cross-sectional spread regressions; proxies time-varying credit risk compensation tied to intermediary constraints. Used with a one-month lag.	Gilchrist and Zakrajšek [2012]
EMERG_DEBT	Excess return on hard-currency emerging market debt.	Brooks et al. [2020]
EMERG_FX	Return on an equal-weighted basket of emerging market currencies versus the USD.	Brooks et al. [2020]
GLOBAL_AGG	Excess return on the Bloomberg Global Aggregate Bond Index.	Brooks et al. [2020]
ITRX_MAIN	First difference of the CDS spread on the iTraxx Europe Main index.	Klaus and Rzepkowski [2009]
MAIN_5/3_FLATTENER	First difference of the natural logarithm of the ratio between the 5-year and 3-year iTraxx Europe Main CDS spreads.	Market indicator
PB_EU_CDS_1Y	Average 1-year CDS spread of European prime brokers.	Klaus and Rzepkowski [2009]
PB_EU_CDS_5Y	Average 5-year CDS spread of European prime brokers.	Klaus and Rzepkowski [2009]
PB_US_CDS_1Y	Average 1-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski [2009]
PB_US_CDS_5Y	Average 5-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski [2009]
RI_EU	Excess return on a European A-rated industrial corporate bond index.	Duarte et al. [2007]
SNR-MAIN	First difference of the 5-year CDS spread difference between the iTraxx Senior Financials index and the iTraxx Europe Main index.	Market indicator
XOVER/MAIN	First difference of the natural logarithm of the ratio between the 5-year iTraxx Crossover and 5-year iTraxx Europe Main CDS spreads.	Market indicator

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 5 (continued)

Factor ID	Description	Reference
	Panel B: Liquidity Factors	
EURIBOR_OIS	Euribor–OIS spread. Difference between 3-month Euribor and the 3-month overnight index swap rate based on EONIA or euroSTR.	Nyborg and Östberg [2014]
GC-REPO_ T-BILL	Spread between the 3-month U.S. Treasury general-collateral repo rate and the 3-month U.S. Treasury bill yield.	Bai and Collin-Dufresne [2019]
HKM_IC	Intermediary capital factor. AR(1) innovation in the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al. [2017]
HPW_NOISE	Treasury noise measure of market-wide illiquidity. Root mean squared deviation between observed U.S. Treasury yields and model-implied yields from a fitted smooth zero-coupon yield curve.	Hu et al. [2013]
ILLIQ	Change in the Amihud illiquidity measure. First difference of the ratio of absolute daily return to daily dollar volume.	Amihud and Mendelson [2015]
LIBOR_OIS	Libor–OIS spread. Difference between 3-month LIBOR and the 3-month overnight index swap rate.	Nyborg and Östberg [2014]
LIBOR_REPO_ SHOCK	AR(2) residual of the spread between 3-month U.S. interbank LIBOR and the 3-month U.S. Treasury general-collateral repo rate.	Asness et al. [2013]
LIQNT	Pastor–Stambaugh non-traded liquidity factor. Innovation in aggregate market liquidity constructed from a cross-sectional average of stock-level order-flow-induced return-reversal measures.	Pástor and Stambaugh [2003]
LIQ_V	Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.	Pástor and Stambaugh [2003]
SILLIQ	Stock illiquidity shock. AR(3) residual of the aggregate Amihud illiquidity measure.	Acharya et al. [2013]
TED_SHOCK_ EU	AR(2) residual of the euro TED spread (3-month Euribor minus the 3-month German government bill rate).	Asness et al. [2013]
TED_SHOCK_ US	AR(2) residual of the U.S. TED spread (3-month LIBOR minus the 3-month U.S. government bill rate).	Asness et al. [2013]
ΔFAILS_PCT_ TSY	First difference of the ratio of monthly total notional amount of U.S. Treasury securities fails-to-deliver reported by primary dealers, excluding TIPS, to U.S. Total Debt Outstanding.	Fleckenstein et al. [2014]
	Panel C: Equity Factors	
BAB_EU	Betting Against Beta (Europe). Long low-beta and short high-beta European equities, each leg rescaled to unit beta.	Frazzini and Pedersen [2014]
BAB_US	Betting Against Beta (US). Long low-beta and short high-beta U.S. equities, each leg rescaled to unit beta.	Frazzini and Pedersen [2014]
CMA_EU	Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios.	Fama and French [2017]
GMOM_EU	Global momentum factor. Return spread between recent winner and loser equity portfolios based on past 12-month returns, expressed in euros.	Asness et al. [2013]
GVAL_EU	Global value factor. Return spread between high and low book-to-market equity portfolios across developed equity regions, expressed in euros.	Asness et al. [2013]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 5 (continued)

Factor ID	Description	Reference
HML_EU	Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios.	Fama and French [2017]
MKT_EU	Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill.	Fama and French [2017]
RB_EU	Excess return on a European bank bond index.	Duarte et al. [2007]
RMW_EU	Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios.	Fama and French [2017]
RS_EU	Excess return on the EuroStoxx Banks index.	Duarte et al. [2007]
SMB_EU	Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios.	Fama and French [2017]
UMD_EU	Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.	Carhart [1997]
	Panel D: Volatility Factors	
ATM_IV_CDX	3-month at-the-money implied volatility on options written on the CDX Investment Grade index.	Novales and Chamizo [2019]
ATM_IV_ITRX	3-month at-the-money implied volatility on options written on the iTraxx Europe Main index.	Novales and Chamizo [2019]
EP_SVIX_1M	Equity premium proxy from the SVIX methodology. Option-implied risk-neutral variance of the market simple return over the next month.	Martin [2017]
EP_SVIX_3M	Same construction as EP_SVIX_1M, using options with a 3-month horizon.	Martin [2017]
IV_BUND	3-month at-the-money implied volatility on the 10-year German Bund futures options.	Cremers et al. [2021]
IV_TSY	3-month at-the-money implied volatility on the 10-year U.S. Treasury futures options.	Cremers et al. [2021]
MOVE	Option-implied U.S. rates volatility. Yield-curve weighted average of at-the-money implied normal yield volatilities from OTC options on 2-, 5-, 10-, and 30-year constant-maturity interest rate swaps; used lagged by one month.	Bansal and Stivers [2022]
PTFSBD	Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.	Fung and Hsieh [2001]
PTFSCOM	Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.	Fung and Hsieh [2001]
PTFSFX	Return on the currency trend-following factor, constructed from lookback straddles on major currency futures.	Fung and Hsieh [2001]
PTFSIR	Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.	Fung and Hsieh [2001]
PTFSSTK	Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.	Fung and Hsieh [2001]
ΔV2X	Change in V2X. Monthly first difference of the V2X index, measuring 30-day implied volatility on Euro STOXX 50 options.	Chung et al. [2019]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 5 (continued)

Factor ID	Description	Reference
ΔVIX	Change in VIX. Monthly first difference of the VIX index, a model-free measure of 30-day implied volatility on S&P 500 options.	Chung et al. [2019]
	Panel E: Macro Factors	
5Y5Y_INFL	5-year-5-year inflation swap rate. Forward-starting EUR inflation swap rate capturing long-term inflation expectations.	Market indicator
BFCI_EU	Bloomberg Euro Area Financial Conditions Index. Composite Z-score of euro-area money, bond, and equity indicators, standardized to the pre-crisis period.	Osina [2019]
EPU_EU	Economic Policy Uncertainty index for Europe. Newspaper-based index aggregated across European countries.	Baker et al. [2016]
EPU_US	Economic Policy Uncertainty index for the United States. Newspaper-based index constructed from normalized frequency of articles referencing economy, uncertainty, and policy.	Baker et al. [2016]
ΔUF	Change in financial uncertainty. First difference of the financial uncertainty index, constructed from a panel of financial market variables.	Ludvigson et al. [2021]
ΔUM	Change in macroeconomic uncertainty. First difference of the macro uncertainty index, a model-based measure of expected volatility in the unforecastable component of a large panel of macroeconomic variables.	Ludvigson et al. [2021]
ΔUR	Change in real uncertainty. First difference of the real-activity uncertainty subindex.	Ludvigson et al. [2021]
	Panel F: Interest Rate Factors	
EURUSD_3M_IV	First difference of the 3-month at-the-money implied volatility on the EUR/USD exchange rate.	Market indicator
GLOBAL_TERM	Excess return on a duration-matched global government bond index.	Brooks et al. [2020]
INFL_LINK	Excess return on global inflation-linked bonds over cash or nominal Treasuries.	Brooks et al. [2020]
R10_EU	Excess return on a European 10-year government bond portfolio.	Duarte et al. [2007]
R2_EU	Excess return on a European 2-year government bond portfolio.	Duarte et al. [2007]
R5_EU	Excess return on a European 5-year government bond portfolio.	Duarte et al. [2007]
SS10Y	Difference between the 10-year EUR fixed swap par rate and the 10-year German Bund yield.	Collin-Dufresn et al. [2001]
SS2Y	Difference between the 2-year EUR fixed swap par rate and the 2-year German Schatz yield.	Collin-Dufresn et al. [2001]
SS5Y	Difference between the 5-year EUR fixed swap par rate and the 5-year German Bobl yield.	Collin-Dufresn et al. [2001]
TERM_EU	Bond term structure factor. Excess total return on long-term German Bunds over the 1-month German government bill return.	Fama and French [1993]
TERM_US	Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.	Fama and French [1993]
YSP_EU	Yield-curve slope factor. Difference between the 5-year and 1-year German government bond yields.	Koijen et al. [2017]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 5 (continued)

Factor ID	Description	Reference
YSP_US	Yield-curve slope factor. Difference between the 5-year and 1-year U.S. Treasury yields.	Koijen et al. [2017]
Δ10Y_YIELD_EU	Monthly change in the 10-year German Bund yield.	Market indicator
Δ10Y_YIELD_US	Monthly change in the 10-year U.S. Treasury yield.	Market indicator
ΔSLOPE_EU	First difference of the euro yield-curve slope (10-year German Bund yield minus 3-month German government bill yield).	Ferson and Harvey [1991]
ΔSLOPE_US	First difference of the U.S. yield-curve slope (10-year Treasury yield minus 3-month Treasury bill yield).	Ferson and Harvey [1991]

A.5 PCA Robustness

A.5.1 Robustness across Window Length and Number of Components

Table 6 reports alpha and $\bar{R}^2$ for all (W,K) combinations. Results are robust to the choice of window length and number of components.

Table 6

PCA Robustness: Alpha and $\bar{R}^2$ Across Window Lengths and Number of Components

W	K	BTP Italia				CDS–Bond				Index Skew				Expl.
		Pred		Cont		Pred		Cont		Pred		Cont		Var.(%)
		α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	
24	8	1.68***	-0.005	1.61***	0.054	5.43***	0.050	5.87***	0.120	2.07***	0.041	2.21***	0.049	80.8
		(4.05)		(3.48)		(5.91)		(5.74)		(5.38)		(5.01)		
24	11	1.65***	0.017	1.70***	0.065	5.42***	0.110	6.02***	0.121	2.10***	0.063	2.19***	0.047	88.5
		(3.98)		(3.67)		(5.60)		(5.74)		(5.26)		(5.00)		
24	14	1.63***	0.058	1.67***	0.071	5.30***	0.170	5.94***	0.120	2.09***	0.066	2.14***	0.127	93.6
		(4.10)		(3.67)		(5.58)		(5.81)		(5.27)		(5.02)		
36	8	1.63***	0.075	1.66***	0.058	5.62***	0.028	5.42***	0.093	2.06***	0.106	1.81***	0.133	76.2
		(3.92)		(3.72)		(5.74)		(5.47)		(4.79)		(4.29)		
36	11	1.72***	0.072	1.62***	0.042	5.72***	0.039	5.50***	0.100	2.08***	0.110	1.80***	0.141	83.5
		(4.06)		(3.80)		(5.85)		(5.59)		(4.91)		(4.22)		
36	14	1.65***	0.066	1.64***	0.030	5.79***	0.046	5.40***	0.117	2.13***	0.117	1.84***	0.133	88.8
		(3.82)		(3.86)		(6.05)		(5.70)		(5.07)		(4.31)		
48	8	1.76***	0.158	1.55***	0.066	4.95***	0.063	5.56***	0.084	1.80***	0.152	1.74***	0.265	73.2
		(4.25)		(3.83)		(4.83)		(5.29)		(4.43)		(4.55)		
48	11	1.82***	0.181	1.55***	0.059	4.87***	0.067	5.55***	0.079	1.78***	0.164	1.70***	0.265	80.5
		(4.59)		(3.72)		(4.74)		(5.46)		(4.29)		(4.44)		
48	14	1.74***	0.176	1.58***	0.063	5.12***	0.071	5.46***	0.076	1.65***	0.179	1.76***	0.269	85.7
		(4.08)		(3.89)		(5.08)		(5.42)		(4.11)		(4.65)		

Note: W is the rolling window length (months); K is the number of retained principal components. Expl. Var. (%) is the average cumulative variance explained by the first K PCs (time-series average across rolling windows). α is annualized (% p.a.). t-statistics in parentheses (Newey–West HAC). Pred $= PC_t \rightarrow R_{t+1}$; Cont $= PC_t \rightarrow R_t$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.5.2 Subperiod and Regime Analysis

Half-sample splits and regime-dependent alpha confirm that PCA-based alpha is stable across subperiods and increases during stress.

Table 7

PCA Subperiod Analysis (Predictive)

	Full Sample	First Half	Second Half	Stress	Normal
BTP Italia					

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(continued)

	Full Sample	First Half	Second Half	Stress	Normal
α (% p.a.)	1.68^{***}	1.48^{***}	1.75^{***}	4.40^{***}	1.41^{***}
	(4.05)	(2.67)	(3.12)	(5.28)	(4.64)
β_{PC1}	-0.012	-0.004	-0.014	-0.024	-0.008
	(-1.13)	(-0.33)	(-1.40)	(-1.39)	(-0.64)
β_{PC2}	-0.001	-0.002	-0.015	-0.078^{**}	0.001
	(-0.07)	(-0.20)	(-1.12)	(-2.09)	(0.07)
β_{PC3}	-0.026	-0.002	-0.040^{*}	-0.109^{**}	-0.024
	(-1.46)	(-0.06)	(-1.68)	(-2.02)	(-1.36)
β_{PC4}	-0.019	0.059	-0.057^{**}	0.048	-0.026
	(-0.61)	(1.38)	(-2.09)	(0.61)	(-0.97)
β_{PC5}	-0.012	-0.041	0.007	-0.316^{***}	0.003
	(-0.52)	(-1.30)	(0.25)	(-4.00)	(0.13)
β_{PC6}	-0.028	-0.043^{*}	-0.021	-0.075	-0.023
	(-1.46)	(-1.69)	(-1.01)	(-0.98)	(-1.37)
β_{PC7}	-0.001	-0.052	0.054	-0.303^{***}	0.029
	(-0.02)	(-1.60)	(1.60)	(-9.76)	(0.95)
β_{PC8}	-0.000	-0.062	0.038^{*}	-0.325^{***}	0.010
	(-0.02)	(-1.30)	(1.85)	(-4.84)	(0.47)
$\bar{R}^2$	-0.005	0.070	0.080	0.419	0.013
N	156	78	78	21	135
CDS–Bond					
α (% p.a.)	5.43^{***}	7.14^{***}	3.58^{***}	7.93^{***}	4.52^{***}
	(5.91)	(4.85)	(4.51)	(3.40)	(4.75)
β_{PC1}	0.018	0.033^{**}	-0.016	0.021	0.003
	(1.23)	(2.02)	(-0.79)	(0.97)	(0.19)
β_{PC2}	0.002	0.014	-0.016	0.020	0.002
	(0.09)	(0.62)	(-0.85)	(0.68)	(0.10)
β_{PC3}	0.068^{*}	0.073	0.048	0.092	0.029
	(1.69)	(1.27)	(1.28)	(1.28)	(0.68)
β_{PC4}	0.059	-0.005	0.090	-0.027	0.091
	(1.17)	(-0.11)	(1.35)	(-0.58)	(1.64)
β_{PC5}	-0.060^{**}	-0.063	-0.060^{*}	-0.052	-0.079^{**}
	(-2.57)	(-1.61)	(-1.70)	(-0.98)	(-2.25)
β_{PC6}	0.011	0.030	-0.014	0.028	0.005
	(0.50)	(0.72)	(-0.54)	(0.42)	(0.20)
β_{PC7}	-0.003	0.042	-0.087^{*}	0.053	-0.050
	(-0.08)	(0.73)	(-1.91)	(0.87)	(-1.20)
β_{PC8}	0.089^{***}	0.105^{***}	0.024	0.113^{***}	0.058
	(3.03)	(3.10)	(0.74)	(2.78)	(1.64)
$\bar{R}^2$	0.050	0.041	0.093	0.026	0.042
N	196	98	98	50	146
Index Skew					
α (% p.a.)	2.07^{***}	2.46^{***}	1.49^{***}	4.35^{***}	1.48^{***}
	(5.38)	(4.81)	(2.83)	(5.67)	(4.50)
β_{PC1}	-0.011	-0.014^{**}	-0.003	-0.001	-0.018
	(-1.20)	(-2.25)	(-0.16)	(-0.09)	(-1.44)

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	Full Sample	First Half	Second Half	Stress	Normal
β_{PC2}	-0.009	-0.008	-0.009	0.019	-0.014
	(-0.95)	(-1.08)	(-0.47)	(1.26)	(-1.28)
β_{PC3}	-0.002	0.022	-0.033	-0.005	-0.007
	(-0.13)	(1.24)	(-1.20)	(-0.12)	(-0.43)
β_{PC4}	0.013	0.019	0.003	0.057**	-0.005
	(0.78)	(1.40)	(0.11)	(1.98)	(-0.25)
β_{PC5}	0.023	0.029*	0.015	0.009	0.033
	(0.89)	(1.77)	(0.32)	(0.23)	(1.11)
β_{PC6}	0.015	-0.010	0.037	-0.015	0.023
	(0.62)	(-0.69)	(1.33)	(-0.44)	(0.97)
β_{PC7}	0.026	-0.003	0.063**	0.029	0.020
	(1.47)	(-0.18)	(2.37)	(0.58)	(1.26)
β_{PC8}	0.042**	0.017	0.074***	-0.012	0.047**
	(2.11)	(0.96)	(2.97)	(-0.50)	(2.37)
$\bar{R}^2$	0.041	0.050	0.053	-0.063	0.094
N	208	104	104	53	155

Note: Stress regime defined as iTraxx Main 5Y > 100 bps. First/Second Half: temporal split at sample midpoint. α is annualized (% p.a.). t-statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8

PCA Subperiod Analysis (Contemporaneous)

	Full Sample	First Half	Second Half	Stress	Normal
BTP Italia					
α (% p.a.)	1.61***	1.82**	1.38*	5.37**	1.28***
	(3.48)	(2.50)	(1.96)	(2.24)	(4.21)
β_{PC1}	−0.025**	−0.034***	-0.017	-0.002	−0.033***
	(-1.99)	(-2.63)	(-0.85)	(-0.07)	(-3.54)
β_{PC2}	-0.010	0.000	-0.018	-0.005	-0.013
	(-0.97)	(0.02)	(-1.16)	(-0.16)	(-1.23)
β_{PC3}	-0.028	-0.041	-0.025	-0.153	-0.016
	(-1.20)	(-1.10)	(-0.78)	(-1.42)	(-0.69)
β_{PC4}	0.017	0.018	0.014	0.009	0.009
	(1.05)	(0.44)	(0.62)	(0.10)	(0.66)
β_{PC5}	0.008	-0.026	0.024	−0.170***	0.021
	(0.41)	(-1.10)	(0.86)	(-2.67)	(1.20)
β_{PC6}	0.003	-0.018	0.009	-0.036	0.000
	(0.13)	(-0.79)	(0.30)	(-0.28)	(0.01)
β_{PC7}	-0.013	-0.037	0.018	−0.210***	0.006
	(-0.50)	(-1.32)	(0.45)	(-3.01)	(0.22)
β_{PC8}	0.010	-0.004	0.021	-0.092	0.005
	(0.60)	(-0.21)	(0.94)	(-1.35)	(0.25)
$\bar{R}^2$	0.054	0.026	0.026	-0.168	0.125
N	157	78	79	21	136
CDS–Bond					

continued on next page

(continued)

	Full Sample	First Half	Second Half	Stress	Normal
α (% p.a.)	5.87***	7.61***	4.08***	9.13***	4.84***
	(5.74)	(4.74)	(3.86)	(3.77)	(4.43)
β_{PC1}	0.035***	0.005	0.051***	0.015	0.049***
	(3.07)	(0.32)	(3.55)	(0.92)	(4.31)
β_{PC2}	0.010	0.013	0.003	0.037	0.008
	(0.52)	(0.51)	(0.10)	(1.08)	(0.36)
β_{PC3}	0.067*	0.074*	0.041	0.076	0.040
	(1.95)	(1.89)	(0.83)	(1.62)	(0.76)
β_{PC4}	-0.016	-0.001	-0.040	-0.008	-0.010
	(-0.54)	(-0.02)	(-1.04)	(-0.16)	(-0.28)
β_{PC5}	-0.032	0.022	−0.071**	0.103*	−0.084**
	(-0.80)	(0.34)	(-2.04)	(1.67)	(-2.15)
β_{PC6}	−0.079**	-0.055	−0.051*	0.016	−0.081**
	(-2.39)	(-1.36)	(-1.70)	(0.19)	(-2.43)
β_{PC7}	0.076**	0.108*	0.048	0.028	0.093**
	(2.02)	(1.71)	(1.04)	(0.41)	(2.00)
β_{PC8}	0.051	0.129*	-0.024	0.075	0.047
	(1.03)	(1.94)	(-0.51)	(0.98)	(0.68)
$\bar{R}^2$	0.120	0.068	0.195	0.026	0.148
N	197	98	99	50	147
Index Skew					
α (% p.a.)	2.21***	2.21***	2.34***	3.38***	1.82***
	(5.01)	(4.60)	(2.80)	(4.36)	(4.22)
β_{PC1}	0.001	0.005	-0.007	-0.001	0.000
	(0.16)	(0.53)	(-0.69)	(-0.09)	(0.02)
β_{PC2}	-0.003	0.003	0.001	0.011	-0.005
	(-0.29)	(0.35)	(0.03)	(0.67)	(-0.43)
β_{PC3}	0.018	-0.016	0.053*	0.020	0.019
	(0.90)	(-1.06)	(1.70)	(0.38)	(0.73)
β_{PC4}	0.042	0.003	0.072**	0.024	0.045
	(1.64)	(0.20)	(2.44)	(1.04)	(1.53)
β_{PC5}	-0.007	-0.016	-0.014	-0.002	-0.013
	(-0.54)	(-1.34)	(-0.85)	(-0.13)	(-1.05)
β_{PC6}	0.012	-0.017	0.013	-0.039	0.020
	(0.65)	(-1.31)	(0.66)	(-0.57)	(1.28)
β_{PC7}	−0.039**	-0.022	−0.080***	-0.053	-0.042
	(-2.19)	(-1.40)	(-3.02)	(-0.91)	(-1.55)
β_{PC8}	0.018	0.018	0.003	0.011	0.004
	(1.21)	(0.92)	(0.12)	(0.37)	(0.25)
$\bar{R}^2$	0.049	-0.006	0.133	-0.104	0.074
N	209	104	105	53	156

Note: Stress regime defined as iTraxx Main 5Y $>$ 100 bps. First/Second Half: temporal split at sample midpoint. α is annualized (% p.a.). t-statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9
PCA Alpha by Regime

	Predictive			Contemporaneous		
Strategy	Full	Stress	Normal	Full	Stress	Normal
BTP Italia	1.68***	4.40***	1.41***	1.61***	5.37**	1.28***
	(4.05)	(5.28)	(4.64)	(3.48)	(2.24)	(4.21)
CDS–Bond	5.43***	7.93***	4.52***	5.87***	9.13***	4.84***
	(5.91)	(3.40)	(4.75)	(5.74)	(3.77)	(4.43)
Index Skew	2.07***	4.35***	1.48***	2.21***	3.38***	1.82***
	(5.38)	(5.67)	(4.50)	(5.01)	(4.36)	(4.22)

Note: Stress regime: iTraxx Main 5Y > 100 bps. α is annualized (% p.a.). t-statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

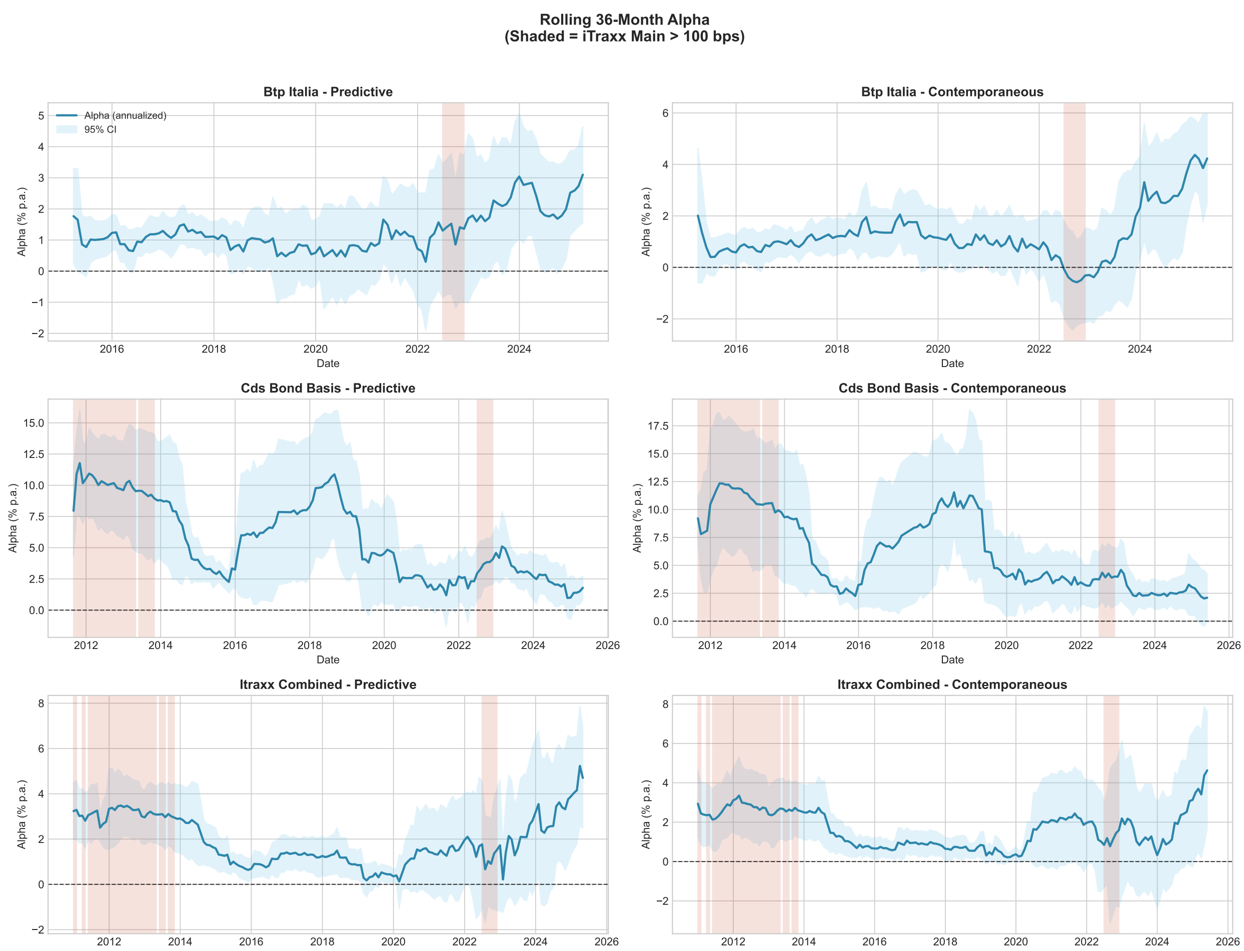

Figure 1
Rolling Alpha with Regime Shading (PCA Factors). Grey bands: 95% confidence intervals. Red shading: iTraxx Main 5Y > 100 bps. Rolling window: 36 months.

A.5.3 Conditional Alpha: Ferson–Schadt and Threshold Robustness (PCA)

Table 10 reports Ferson–Schadt conditional alpha using PCA factors. Table 11 confirms robustness across stress thresholds (80, 100, 120 bps).

Table 10
Conditional Alpha (PCA, Contemporaneous): Ferson–Schadt (1996) Model

	BTP Italia	CDS–Bond Basis	iTraxx Combined
α_0 (ann. %)	$+1.61^{***}$	$+5.87^{***}$	$+2.21^{***}$
α_1 (conditional)	$+0.1168^{***}$	$+0.2292^{***}$	$+0.0736^{**}$
t-stat	(3.08)	(2.77)	(2.54)
Joint F-test	$+7.06^{***}$	$+3.18^{***}$	$+3.36^{***}$
$\alpha(z=+1\sigma)$ ann.	+3.17%	+8.58%	+3.09%
$\alpha(z=-1\sigma)$ ann.	+0.36%	+3.08%	+1.32%
T	157	197	209

Note: Ferson and Schadt (1996) conditional model with rolling PCA factors. Timing: Contemporaneous ($PC_t \to R_{t+1}$ if predictive). z_t = standardized iTraxx Main 5Y. t-statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11
Conditional Alpha (PCA, Contemporaneous): Stress Threshold Robustness

		BTP Italia			CDS–Bond Basis			iTraxx Combined		
Threshold	n_H	α_1 (ann.)	t	p	α_1 (ann.)	t	p	α_1 (ann.)	t	p
80 bps	42	+1.48%	1.17	0.2424	$+4.36^{***}$%	2.62	0.0087	$+1.36^{*}$%	1.78	0.0750
100 bps	21	+2.37%	1.14	0.2550	+3.95%	1.63	0.1033	+1.31%	1.56	0.1198
120 bps	10	$+6.77^{**}$%	2.25	0.0242	+5.16%	1.58	0.1148	$+3.05^{**}$%	2.57	0.0103

Note: $D_{\text{HIGH}} = \mathbf{1}\{\text{iTraxx Main 5Y} > \text{threshold}\}$. PCA rolling factors. t-statistics (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.5.4 Variance Diagnostics

Figure 2 shows the time-varying cumulative variance explained.

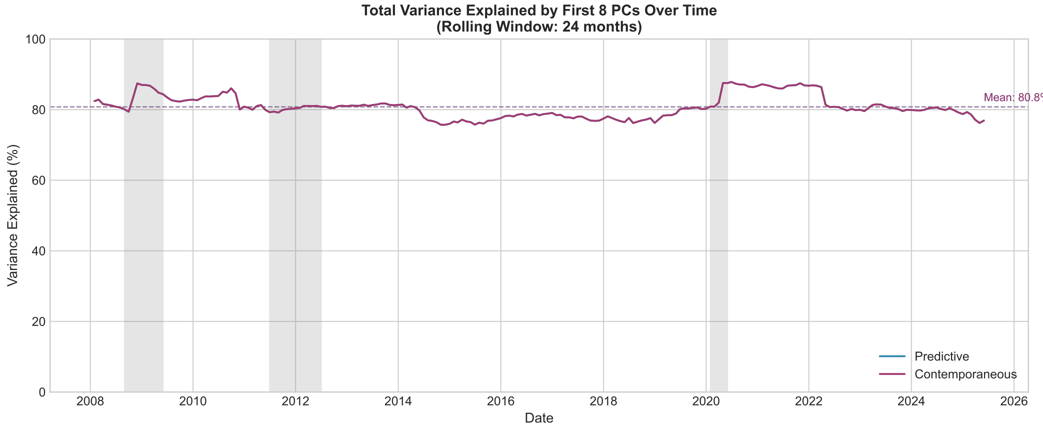

Figure 2

Time Series of Cumulative Variance Explained by Rolling PCA.

A.6 AEN Robustness

A.6.1 Sensitivity to Hyperparameters

We verify that factor selection is stable to the correlation pre-cleaning threshold ($\rho \in \{0.85, 0.90, 0.95\}$) and the adaptive weight exponent ($\gamma \in \{1, 2, 3\}$). The AEN selects the same set of factors across all configurations, with the post-selection alpha essentially unchanged. No selected factor belongs to a pair correlated above the baseline threshold, and the Stage 1 elastic net coefficients exhibit sufficient separation between relevant and irrelevant factors to make the adaptive weighting insensitive to the exponent.

A.6.2 Factor Selection Across Methods

Table 12 reports the factors selected by each penalised method for each strategy. LASSO, Elastic Net, and Adaptive LASSO produce broadly overlapping selections, while AEN—which retains only factors exceeding the 80% bootstrap stability threshold—yields the sparsest models. ALASSO-Chen selects substantially more factors across all strategies, reflecting its use of AIC for tuning λ_1 [Chen et al., 2025], a less stringent criterion than the HQC adopted for the baseline AEN.

Table 12

Selected Factors by Method and Strategy

Method	**BTP Italia**	**CDS–Bond Basis**	**iTraxx Combined**
LASSO	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX, LIBOR_OIS, SMB_EU, ΔVIX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY, MOVE, LIBOR_REPO_SHOCK
Elastic Net	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX, LIBOR_OIS, SMB_EU, ΔVIX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY, MOVE, LIBOR_REPO_SHOCK
Ada. LASSO	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY

(continued)

Method	**BTP Italia**	**CDS–Bond Basis**	**iTraxx Combined**
AEN	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY
ALASSO-Chen	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, TERM_EU, TERM_US, ATM_IV_CDX, ΔSLOPE_EU, Δ10Y_YIELD_US, LIBOR_REPO_SHOCK, ΔSLOPE_US, TED_SHOCK_US, ΔFAILS_PCT_TSY, DEF_US, CREDIT_EU, EPU_EU, ΔUF, YSP_US, EPU_US, MKT_EU, BAB_US, BAB_EU, PB_EU_CDS_1Y, PB_US_CDS_1Y, PB_EU_CDS_5Y, HPW_NOISE, GC-REPO_T-BILL, GMOM_EU, SILLIQ, SNR-MAIN, MAIN_5/3_FLATTENER, ITRX_MAIN, XOVER/MAIN, RMW_EU, HML_EU, SS5Y, BFCI_EU, IV_BUND, SS2Y, EURUSD_3M_IV, PTFSBD, PTFSCOM, 5Y5Y_INFL, PTFSSTK, R5_EU, RI_EU, R10_EU, GLOBAL_AGG, CORP_CREDIT, EMERG_DEBT, EMERG_FX	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, LIBOR_OIS, SMB_EU, CREDIT_US, DEF_US, EP_SVIX_3M, TERM_US, ΔV2X, CRED_SPR_EU, EPU_US, MKT_EU, BAB_US, TED_SHOCK_EU, PB_US_CDS_1Y, LIBOR_REPO_SHOCK, HML_EU, HPW_NOISE, CMA_EU, SS10Y, XOVER/MAIN, MOVE, EBP, PTFSBD, EURUSD_3M_IV, EURIBOR_OIS, GLOBAL_AGG	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HKM_IC, EP_SVIX_3M, TERM_US, DEF_EU, ΔSLOPE_EU, TED_SHOCK_EU, PB_EU_CDS_5Y, CDX_IG, SS5Y, EMERG_DEBT

Note: Each cell lists the factors selected (non-zero coefficient) by the corresponding method. Ridge excluded (selects all). AEN denotes the Adaptive Elastic Net with bootstrap stability selection ($\hat{\pi}_j \geq 60\%$).

A.6.3 Conditional Alpha: Detailed Results (AEN)

Table 13 reports the full Ferson–Schadt conditional alpha model with AEN-selected factors. Table 14 confirms robustness across stress thresholds (80, 100, 120 bps).

Table 13

Conditional Alpha: Ferson–Schadt (1996) Model

	BTP Italia	CDS–Bond Basis	iTraxx Combined
Panel A: Conditional alpha			
α_0 (ann. %)	$+1.39^{***}$	$+3.94^{***}$	$+0.47$
α_1 (conditional)	$+0.0428$	$+0.2352^{**}$	$+0.0471$
t-stat	(1.31)	(2.51)	(1.55)
Joint F-test	3.27^{***}	3.76^{***}	2.17^{**}
$\bar{R}^2$ (unconditional)	0.2173	0.2670	0.2359
$\bar{R}^2$ (conditional)	0.2778	0.2997	0.2533
$\alpha(z=+1\sigma)$ ann.	+1.67%	+7.83%	+0.87%
$\alpha(z=-1\sigma)$ ann.	+0.65%	+2.19%	-0.26%
T	157	197	237
Panel B: Conditional betas (δ_j)			
$\text{CDX}_{\text{IG}} \times z_t$	-0.002	–	–
$\text{CRED}_{\text{SPR\_US}} \times z_t$	–	$+0.215$	–
$\text{EMERG}_{\text{FX}} \times z_t$	–	$+0.016$	–
$\text{EP}_{\text{SVIX\_1M}} \times z_t$	–	$+0.175$	-0.249
$\text{ILLIQ} \times z_t$	$+0.378$	–	–
$\text{LIBOR}_{\text{OIS}} \times z_t$	$+0.099$	–	-0.379^{*}
$\text{PB}_{\text{EU\_CDS\_1Y}} \times z_t$	–	-0.003	–
$\text{PTFSFX} \times z_t$	–	$+0.007$	–
$\text{RI}_{\text{EU}} \times z_t$	–	-0.012	–
$\text{SMB}_{\text{EU}} \times z_t$	-0.032^{***}	–	–
$\text{SS10Y} \times z_t$	$+0.006$	–	–
$\text{SS5Y} \times z_t$	–	$+0.015^{**}$	–
$\text{UMD}_{\text{EU}} \times z_t$	$+0.017$	–	–
$\Delta\text{FAILS}_{\text{PCT\_TSY}} \times z_t$	–	–	$+0.314^{*}$
$\Delta\text{UF} \times z_t$	–	-0.541	–
$\Delta\text{UR} \times z_t$	–	–	$+1.690$
$\Delta\text{V2X} \times z_t$	–	–	$+0.003$

Note: Ferson and Schadt (1996) conditional performance model. z_t is the standardized iTraxx Main 5Y spread (mean zero, unit variance). $\alpha_1 > 0$ indicates alpha increases with credit stress. Joint F-test: H_0: $\alpha_1 = \delta_1 = \cdots = \delta_k = 0$. HAC (Newey–West) standard errors with 6 lags. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Factor definitions are provided in Table 5.

Table 14

Conditional Alpha: Stress Threshold Robustness

	BTP Italia			CDS–Bond Basis			iTraxx Combined		
Threshold	n_H	α_1 (ann.)	t-stat	n_H	α_1 (ann.)	t-stat	n_H	α_1 (ann.)	t-stat
80 bps	42	+0.67%	0.68	82	+2.38%	1.60	88	+0.48%	0.73
100 bps	21	+1.55%	0.99	50	$+3.15^{*}$%	1.71	53	+0.24%	0.30
120 bps	10	+4.19%	1.40	27	$+5.32^{*}$%	1.89	28	+1.37%	0.98

Note: $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH},t} + \sum_j \beta_j X_{jt} + \varepsilon_t$, where $D_{\text{HIGH}} = \mathbf{1}\{\text{iTraxx Main 5Y} > \text{threshold}\}$. n_H = number of HIGH months. Factors from bootstrap stability selection. HAC (Newey–West) standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.6.4 Rolling Alpha

Figure 3 plot rolling 36-month alpha with regime shading for AEN selected factors. Alpha spikes during stress episodes.

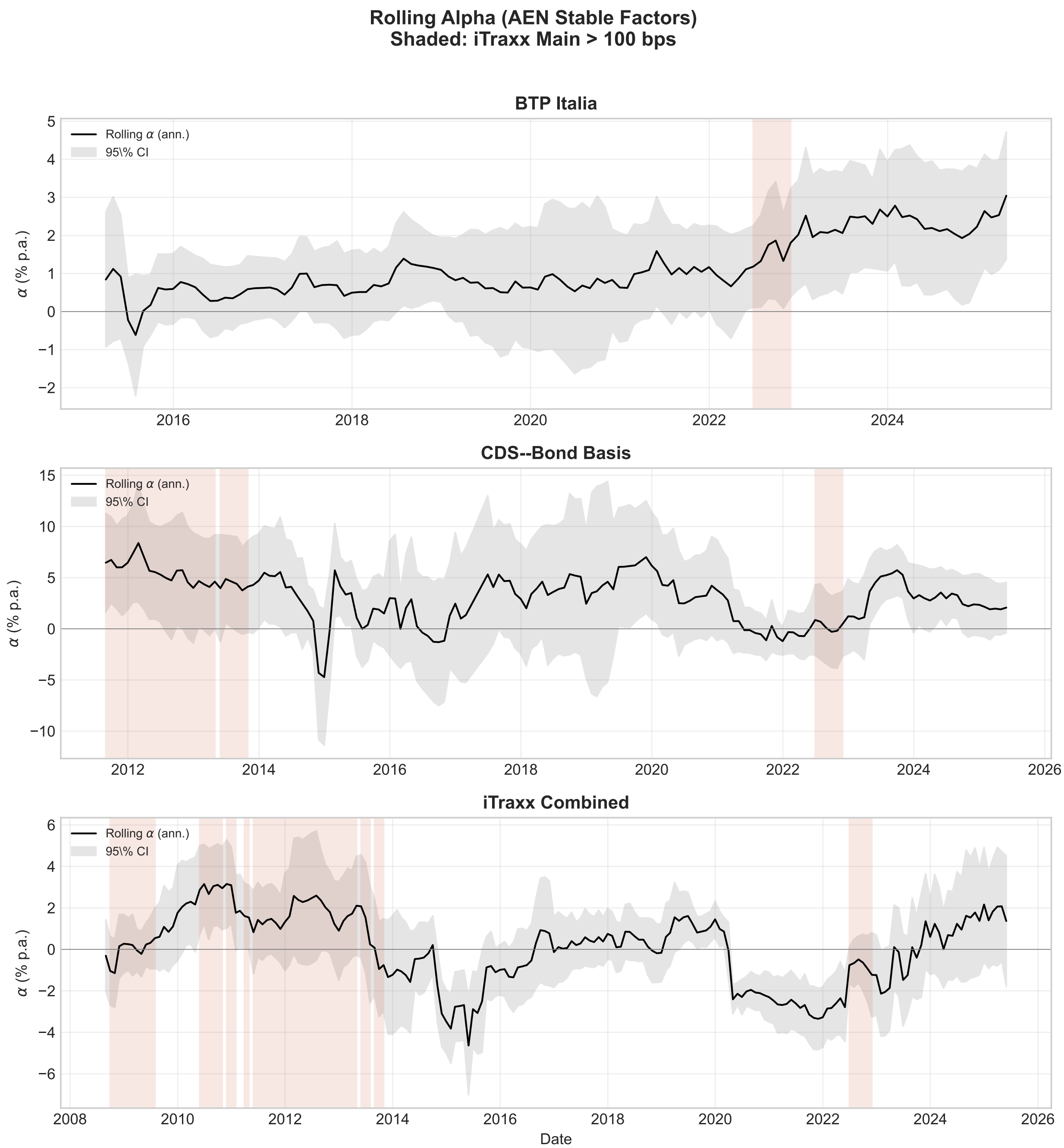

Figure 3
Rolling Alpha with Regime Shading (AEN Stable Factors). Grey bands: 95% confidence intervals. Red shading: iTraxx Main 5Y $>$ 100 bps. Rolling window: 36 months.

A.7 Subperiod and Rolling Window Analysis

This appendix reports sub-period splits and threshold robustness for the benchmark factor models of Section 3. Each table contains Panel A (full sample, first half, second half, HIGH, and NORMAL regimes) and Panel B (stress threshold robustness at 80, 100, and 120 bps of iTraxx Main 5Y).

A.7.1 Duarte et al. (2007)

Table 15 reports sub-period splits and stress regime alpha for the Duarte et al. [2007] specification.

Table 15

Sub-Period and Regime Analysis: Duarte et al. (2007)

Panel A: Sub-period alpha (% p.a.)

Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.65***	3.46	162	+2.26***	4.58	241	+5.14***	5.08	205
First Half	+1.11**	1.97	81	+2.04***	4.25	120	+7.07***	4.37	102
Second Half	+1.79**	2.35	81	+3.03***	3.64	121	+2.89***	3.49	103
HIGH	+4.74**	2.17	21	+2.71***	3.32	53	+8.63***	5.70	50
NORMAL	+1.65***	3.29	141	+1.98***	3.05	188	+3.77***	3.38	155

Panel B: Alpha by stress threshold (% p.a.)

Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis	
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL
80 bps	42	+2.64	+1.30	+3.39***	+1.53	+7.30**	+3.51
100 bps	21	+3.09	+1.46	+3.12	+2.00	+8.67**	+3.90
120 bps	10	+5.80	+1.48	+4.24**	+1.97	+10.45**	+4.24

Notes: Factor model: Duarte et al. (2007). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t-test on α_1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

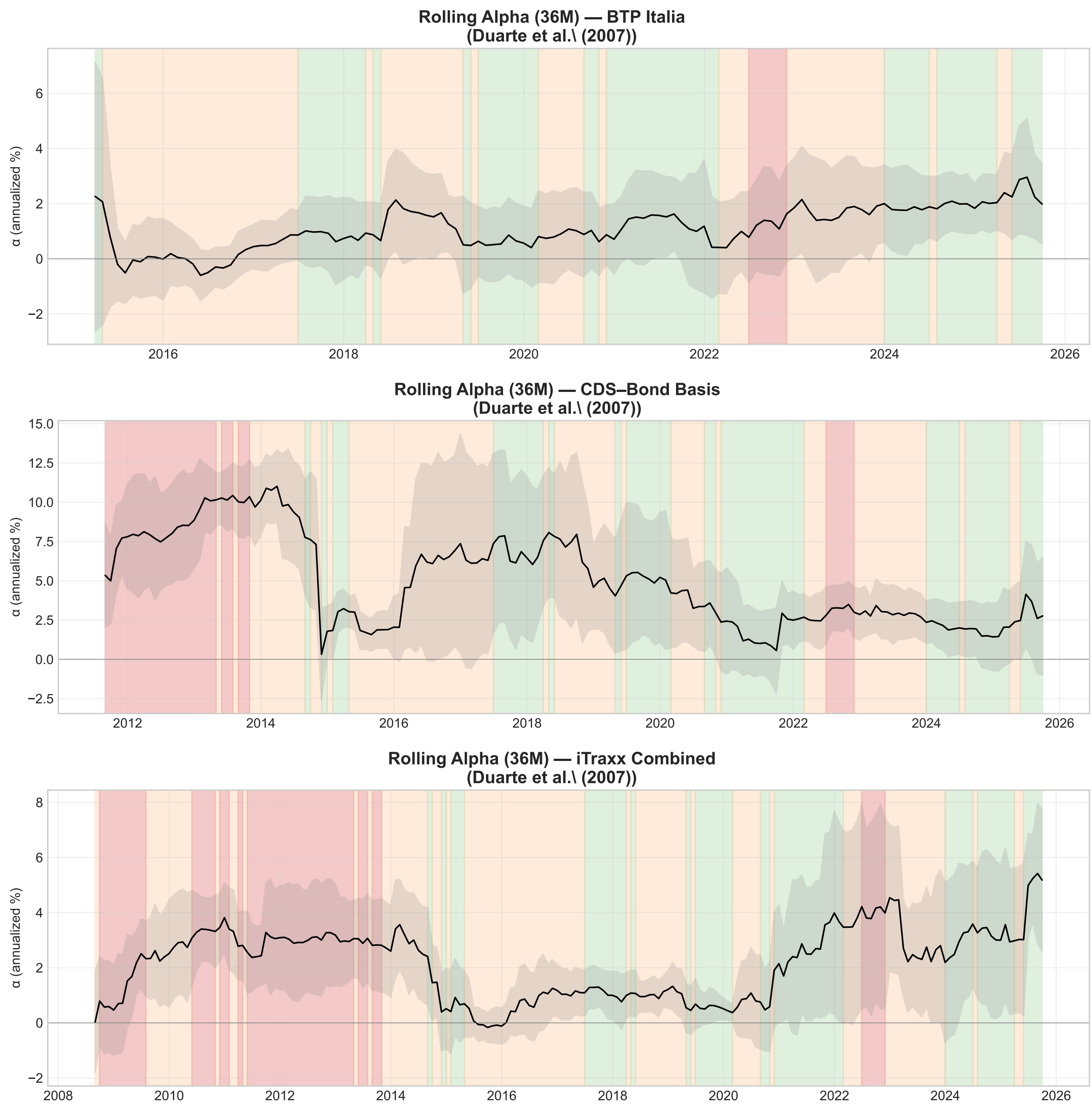

Figure 4
Rolling 36-Month Alpha with Regime Shading — Duarte et al. (2007). 95% confidence bands. Red: HIGH stress (iTraxx Main > 100 bps).

A.7.2 Brooks et al. (2020) — Active Fixed Income

Table 16 reports the same analysis under the Brooks et al. [2020] specification.

Table 16

Sub-Period and Regime Analysis: Active FI (Brooks et al. (2020))

Panel A: Sub-period alpha (% p.a.)

Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.93***	3.19	163	+1.96***	4.50	243	+5.07***	5.28	207
First Half	+2.28***	2.81	81	+2.04***	4.21	121	+7.64***	4.94	103
Second Half	+1.09*	1.83	82	+2.08***	2.61	122	+2.31***	2.81	104
HIGH	+4.13**	2.15	21	+4.38***	4.48	53	+9.69***	4.80	50
NORMAL	+1.42***	3.58	142	+1.38***	3.01	190	+3.63***	4.42	157

Panel B: Alpha by stress threshold (% p.a.)

Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis	
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL
80 bps	42	+3.47*	+1.44	+3.25**	+1.30	+7.94***	+3.36
100 bps	21	+3.99	+1.73	+3.37**	+1.59	+9.22***	+3.82
120 bps	10	+7.99**	+1.65	+5.07***	+1.59	+10.87**	+4.27

Notes: Factor model: Active FI (Brooks et al. (2020)). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t-test on α_1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

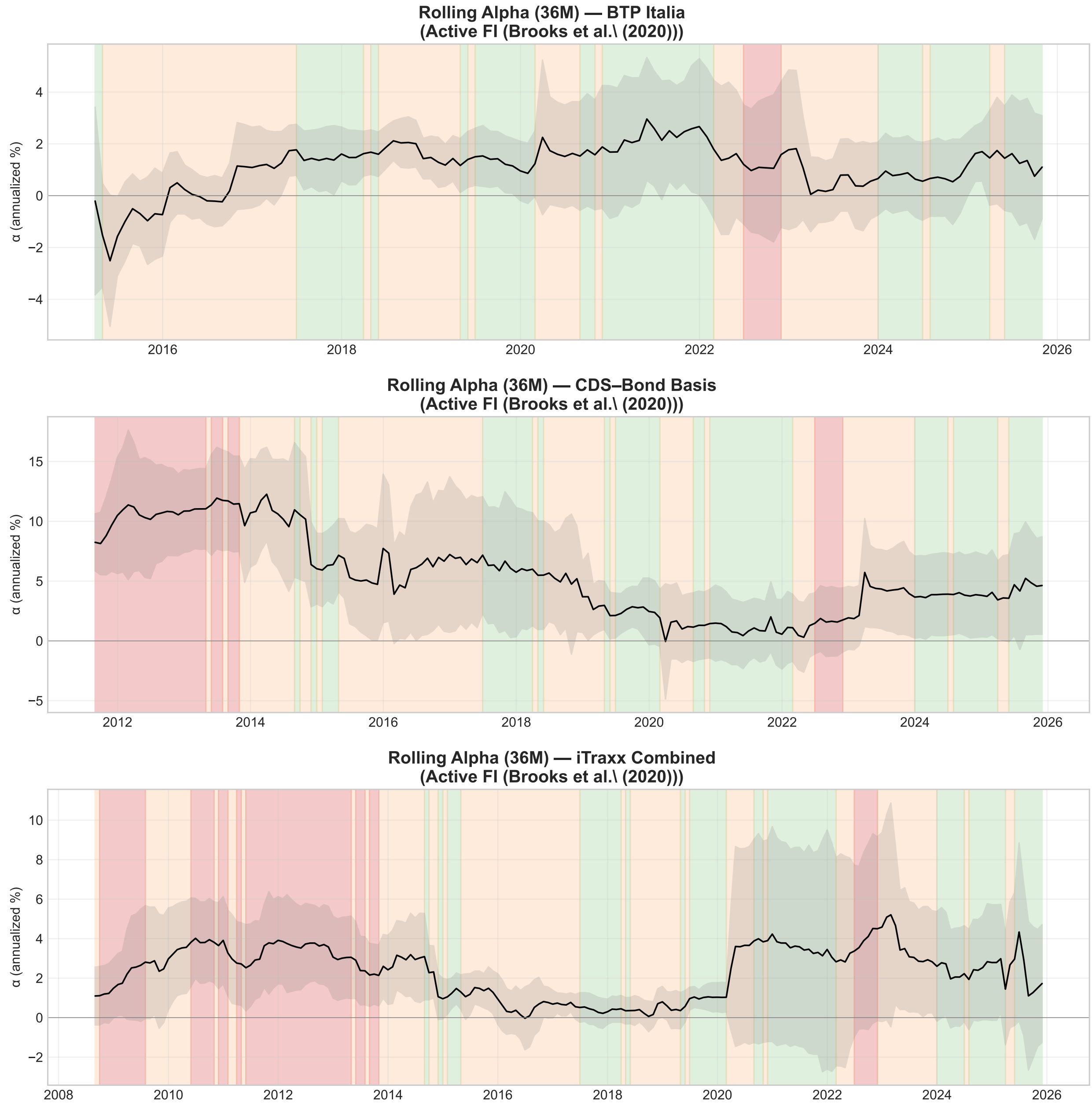

Figure 5

Rolling 36-Month Alpha with Regime Shading — Brooks et al. (2020). Same format as Figure 4.

A.7.3 Fung & Hsieh (2004)

Table 17 reports results for the Fung and Hsieh [2004] specification.

Table 17
Sub-Period and Regime Analysis: Fung & Hsieh (2001)

Panel A: Sub-period alpha (% p.a.)

Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.86***	3.54	162	+3.48***	5.31	195	+4.67***	4.74	195
First Half	+2.05**	2.49	81	+5.00***	4.64	97	+7.25***	4.89	97
Second Half	+1.46**	2.21	81	+2.45***	3.88	98	+2.45***	3.59	98
HIGH	+5.62***	3.80	21	+7.25***	4.71	40	+9.56***	5.14	40
NORMAL	+1.46***	3.61	141	+2.61***	4.91	155	+2.95***	4.89	155

Panel B: Alpha by stress threshold (% p.a.)

Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis	
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL
80 bps	42	+2.76	+1.52	+6.46***	+1.58	+7.49***	+2.87
100 bps	21	+4.43	+1.49	+7.21***	+2.52	+10.27**	+3.23
120 bps	10	+7.88**	+1.44	+11.71***	+2.56	+13.04**	+3.73

Notes: Factor model: Fung & Hsieh (2001). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t-test on α_1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

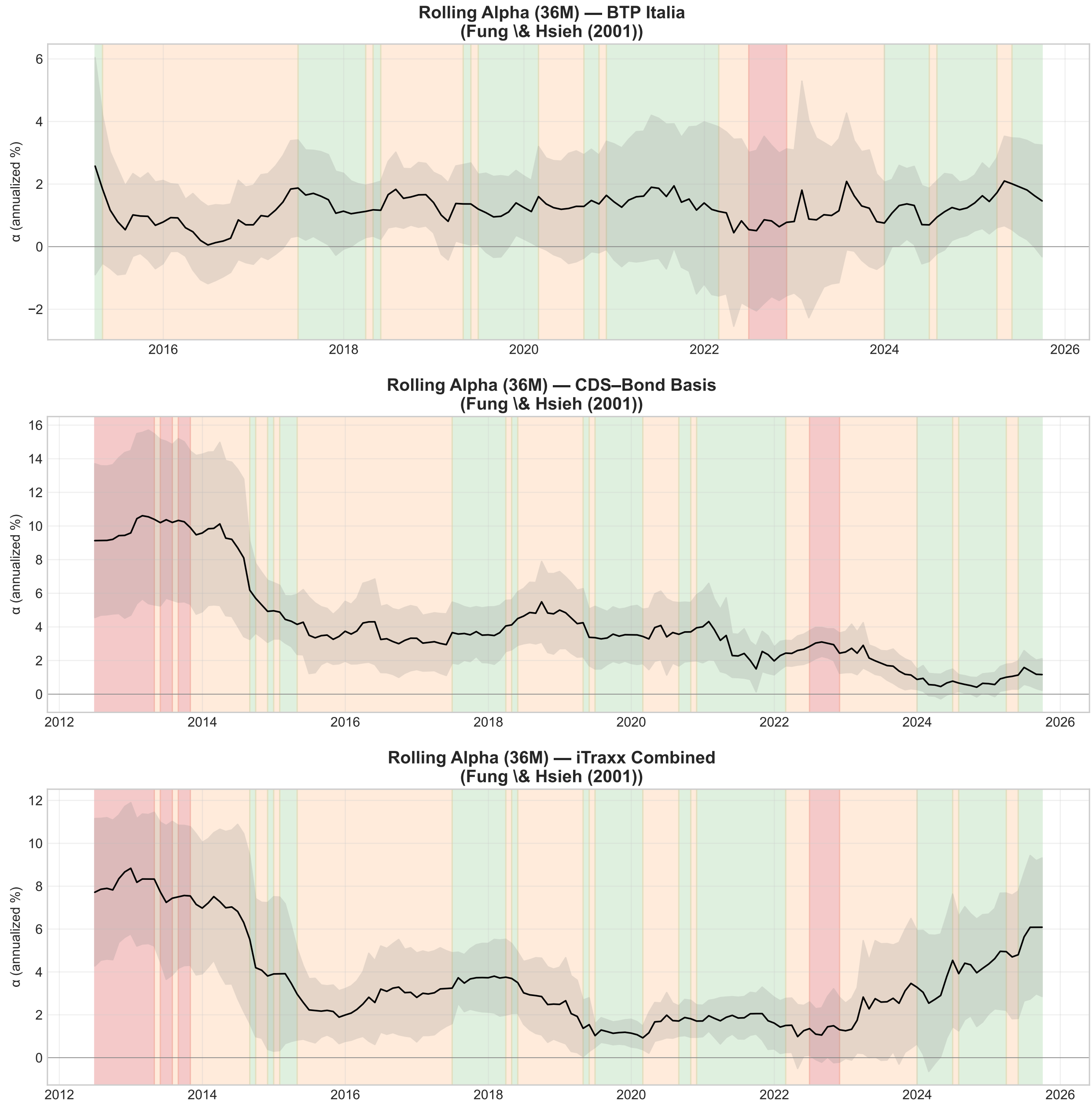

Figure 6
Rolling 36-Month Alpha with Regime Shading — Fung & Hsieh (2004). Same format as Figure 4.

A.8 Cross-Strategy Interdependencies: Additional Tests

This appendix collects robustness checks and supplementary analyses for Section 5. All tests use the same mispricing magnitudes $m_{i,t}$ and regime definitions as the main text.

A.8.1 Purged Correlations

To verify that cross-strategy co-movement is not driven solely by common risk-factor exposures, we regress each Δm_i on the AEN-selected factors from Section 4 and compute correlations on the residuals. Table 18 shows that pairwise correlations remain positive and significant after purging, indicating that the interdependencies documented in Section 5 reflect capital-market linkages beyond shared factor exposures.

Table 18
Correlations After Purging Common Factors

Pair	Raw	Purged	Reduction
CDS–Bond Basis vs BTP Italia	-0.260	-0.272	-4.87%
CDS–Bond Basis vs iTraxx Combined	0.182	0.130	28.87%
BTP Italia vs iTraxx Combined	-0.152	-0.093	38.69%

Note: Correlations of residuals after regressing each strategy's Δm on common factors (EURIBOR-OIS, HKM intermediary capital, HPW noise). Reduction shows the percentage decline in absolute correlation.

A.8.2 DCC-GARCH

We estimate bivariate DCC-GARCH(1,1) models on each pair of Δm series. Figure 7 plots the time-varying conditional correlations. Spikes in conditional correlation coincide with stress episodes (COVID-19, ECB hiking cycle), corroborating the regime-dependent results in Section 5.

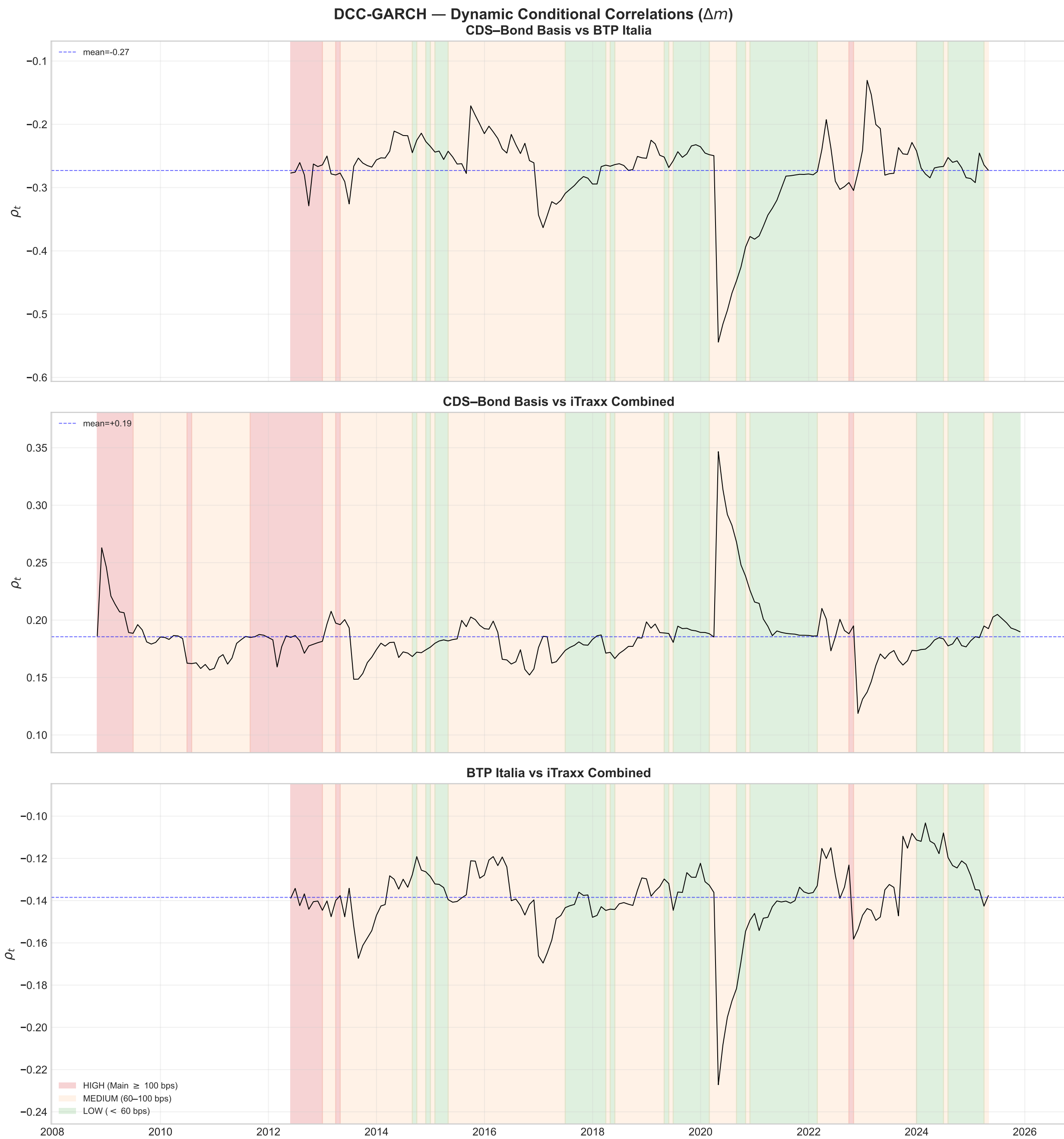

Figure 7
Dynamic Conditional Correlations (DCC-GARCH). Bivariate DCC(1,1) estimates for each pair of Δm. Shaded areas: HIGH stress regime (iTraxx Main > 100 bps).

A.8.3 Granger Causality: Lag Robustness

Table 19 reports Granger causality tests at lag lengths $p = 1, 2, 3$. The main result — iTraxx Combined Granger-causes BTP Italia — is robust across all specifications.

Table 19
Granger Causality: Robustness to Lag Length

Cause	Effect	$p=1$		$p=2$		$p=3$	
		F	p-val	F	p-val	F	p-val
BTP Italia	CDS–Bond Basis	0.81	0.3696	0.37	0.6879	0.51	0.6769
iTraxx Combined	CDS–Bond Basis	2.34	0.1265	3.03**	0.0495	1.70	0.1660
CDS–Bond Basis	BTP Italia	3.97**	0.0468	1.67	0.1897	0.90	0.4414
iTraxx Combined	BTP Italia	9.51***	0.0022	4.61**	0.0104	2.99**	0.0310
CDS–Bond Basis	iTraxx Combined	0.02	0.8922	0.30	0.7434	0.29	0.8346
BTP Italia	iTraxx Combined	3.14*	0.0771	2.55*	0.0795	1.58	0.1934

Note: Granger causality F-tests from VAR(p) on Δm, estimated separately at $p = 1,2,3$ lags. Lag selection: HQC selects $p = 1$; AIC/FPE select $p = 2$; BIC selects $p = 0$ (forced to 1). $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.10$.

A.8.4 Alternative Stress Proxies

Table 20 replicates key tests from Section 5 (unconditional correlations, regime correlations, interaction regressions) using iTraxx Crossover, V2X, and VIX as alternative conditioning variables instead of iTraxx Main. Results are qualitatively unchanged, confirming that the documented interdependencies are not an artifact of the stress-proxy choice.

Table 20
Robustness: Alternative Stress Proxies

Pair	iTraxx Main		iTraxx Xover		V2X		VIX	
	ρ_H	ρ_L	ρ_H	ρ_L	ρ_H	ρ_L	ρ_H	ρ_L
CDS–Bond Basis vs BTP Italia	-0.54	-0.08	-0.36	-0.10	-0.65	-0.04	-0.72	-0.02
CDS–Bond Basis vs iTraxx Combined	-0.25	-0.12	-0.09	+0.25	-0.94	-0.12	-0.08	-0.01
BTP Italia vs iTraxx Combined	+0.50	+0.20	+0.24	-0.12	+0.78	+0.17	+0.36	+0.17

Note: ρ_H and ρ_L are Pearson correlations of strategy returns in HIGH and LOW stress regimes, respectively. Regimes defined by manual thresholds on each proxy. Sign patterns are consistent across all proxies, confirming that cross-strategy interdependencies are not an artifact of the conditioning variable.

A.9 VIF Diagnostics

Tables 21–23 report Variance Inflation Factors for the EUR factor specifications used in the benchmark regressions of Section 3. VIF is computed on the factor matrix for each strategy's sample period; because the factors are identical across strategies, differences in VIF reflect only the different estimation windows.

All equity and bond-market factors exhibit VIF below 10, confirming acceptable multicollinearity levels. The Duarte specification shows elevated VIF for the term-structure factors R2, R5, and R10 (VIF $\approx$ 15–55), which is expected given the high correlation structure of yield-curve returns and does not affect inference on alpha. The Active FI and Fung & Hsieh specifications show VIF below 5 for all factors.

Table 21
Variance Inflation Factors — Duarte et al. (2007), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
Mkt-RF	5.02	5.37	5.06
SMB	1.14	1.25	1.19
HML	3.10	3.13	2.99
UMD	1.50	1.78	1.72
RS	7.25	7.96	7.90
RI	16.03	5.43	5.24
RB	11.84	2.83	2.79
R2	15.17	13.97	15.41
R5	54.39	50.88	54.08
R10	26.65	24.18	23.86

Note: VIF computed on EUR factor matrix. VIF $>$ 10 indicates serious multicollinearity; VIF $>$ 5 moderate concerns. Elevated VIF for R2, R5, R10 reflects the inherent correlation of Treasury return factors and does not affect inference on alpha.

Table 22
Variance Inflation Factors — Brooks et al. (2020), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
Term	6.48	6.49	6.46
Global_Term	50.36	41.58	40.74
Global_Aggregate	46.25	38.52	37.66
Inflation_Linkers	4.70	3.73	3.54
Corporate_Credit	2.49	3.51	3.55
Emerging_Debt	3.17	5.10	4.64
Emerging_Currency	1.30	1.22	1.27
UST_Volatility	1.11	1.10	1.08

Note: VIF computed on EUR factor matrix. VIF $>$ 10 indicates serious multicollinearity; VIF $>$ 5 moderate concerns.

Table 23
Variance Inflation Factors — Fung & Hsieh (2004), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
SNP	1.92	2.05	2.05
SIZE	1.05	1.05	1.05
PTFSBD	1.42	1.42	1.42
PTFSFX	1.45	1.44	1.44
PTFSCOM	1.19	1.22	1.22
TERM	1.03	1.01	1.01
CREDIT	1.89	1.98	1.98

Note: VIF computed on EUR factor matrix. VIF $>$ 10 indicates serious multicollinearity; VIF $>$ 5 moderate concerns.

Tables

Table 24

Summary Statistics for Fixed Income Arbitrage Strategies

Panel A: Basis Levels (bps)

Strategy	Mean	Median	Std Dev	Min	Max	Skew	Kurt	AC(1)	P_{10}	P_{90}
BTP Italia	45.55	48.70	29.34	-83.9	119.3	-1.37	3.03	0.984	14.8	75.6
CDS-Bond Basis	7.03	10.06	18.44	-129.7	41.2	-1.01	1.90	0.944	-18.8	29.6
iTraxx Combined	0.40	1.05	6.45	-43.8	24.8	-1.74	8.42	0.872	-5.9	6.6

Panel B: Strategy Returns (%)

Strategy	n	Mean (p.a.)	Std Dev (p.a.)	Min	Max	Skew	Kurt	% Neg	AC(1)	Sharpe
BTP Italia	3282	1.67	2.36	-1.60	1.77	0.60	22.85	48.9	-0.031	0.71
CDS Bond Basis	4169	5.37	3.15	-1.56	3.62	2.81	46.91	43.2	-0.029	1.71
iTraxx Combined	5018	1.96	2.58	-1.03	1.05	0.43	5.67	48.8	-0.231	0.76

Note: Panel A reports daily basis levels in basis points (Jan 2005–May 2025 for CDS–Bond, Oct 2008–May 2025 for iTraxx, Nov 2013–May 2025 for BTP Italia). For BTP Italia, the basis is the cross-sectional mean of all outstanding issues with at least six months to maturity. For iTraxx Combined, it is the mean of the four on-the-run sub-index skews (Main, SeniorFin, SubFin, Crossover). For CDS–Bond Basis, it is the cross-sectional median of all single-name bases in the iTraxx Main universe. P_{10} and P_{90} are the 10th and 90th percentiles of the daily time series. Panel B reports daily percentage returns, with *Mean* and *Std Dev* annualized ($\times 252$ and $\times\sqrt{252}$). *Skew* and *Kurt* denote skewness and excess kurtosis. *AC(1)* is the first-order autocorrelation. *Sharpe* is annualized. Sample periods vary by strategy.

Table 25
Manipulation-Proof Performance Measure by Risk Aversion

Strategy	$\rho = 2$ (% p.a.)	$\rho = 3$ (% p.a.)	$\rho = 4$ (% p.a.)	Avg Ret (% p.a.)	Vol (% p.a.)	Sharpe	N
Panel A: EW							
BTP-Italia	1.31	1.30	1.28	1.34	1.69	0.79	163
CDS-Bond Basis	4.28	4.25	4.21	4.36	2.71	1.61	207
iTraxx *combined*	1.31	1.31	1.30	1.33	1.04	1.27	243
Panel B: SW							
BTP-Italia	1.56	1.55	1.54	1.59	1.62	0.98	163
CDS-Bond Basis	5.06	5.01	4.97	5.16	3.07	1.68	207
iTraxx *combined*	1.83	1.82	1.80	1.85	1.52	1.22	243

Note: MPPM is the Manipulation-Proof Performance Measure (Goetzmann et al. 2007). Computed from monthly returns compounded from daily. Risk-free is Euribor 1M (ACT/360), compounded to monthly.

Table 26
Moreira-Muir Volatility-Managed Portfolio Performance

Strategy	Buy-and-Hold			Vol-Managed			Alpha	N
	Ret (% p.a.)	Vol (% p.a.)	Sharpe	Ret (% p.a.)	Vol (% p.a.)	Sharpe	(% p.a.)	
Panel A: EW								
BTP-Italia	1.27	1.67	0.76	0.21	1.67	0.13	-0.33	158
CDS-Bond Basis	4.27	2.63	1.63	2.83	2.63	1.08	+1.17	202
iTraxx *combined*	1.36	1.05	1.29	0.60	1.05	0.57	-0.01	242
Panel B: SW								
BTP-Italia	1.52	1.60	0.95	0.37	1.60	0.23	-0.22	158
CDS-Bond Basis	5.09	3.00	1.70	3.49	3.00	1.16	+1.68	202
iTraxx *combined*	1.91	1.53	1.25	0.84	1.53	0.55	-0.16	242

Note: Following Moreira and Muir (2017). The volatility-managed strategy scales positions inversely to realized variance: $f^{\sigma}_{t+1} = (c/\sigma_t^2) \times f_{t+1}$, where c matches buy-and-hold volatility. Alpha is from regressing VM returns on BH returns.

Table 27

Duarte Factor Model Regressions

	α (%)	β_{Mkt}	β_{SMB}	β_{HML}	β_{UMD}	β_{R_S}	β_{R_I}	β_{R_B}	β_{R_2}	β_{R_5}	$\beta_{R_{10}}$	$\bar{R}^2$
Panel A: US Factors												
BTP Italia	2.05^{***}	-0.07^{***}	-0.03^{**}	-0.04^{**}	0.00	0.04^{***}	0.04	-0.08	0.09	-0.05	0.04	0.13
	(4.41)	(-3.40)	(-2.19)	(-2.44)	(0.07)	(3.40)	(0.55)	(-0.94)	(0.35)	(-0.31)	(0.54)	
CDS–Bond	5.13^{***}	-0.01	0.00	0.01	-0.03^{*}	-0.01	0.26^{***}	-0.06	0.11	-0.46^{*}	0.12	0.17
	(6.66)	(-0.43)	(0.08)	(0.40)	(-1.75)	(-0.39)	(3.32)	(-1.59)	(0.31)	(-1.84)	(1.08)	
Index Skew	1.81^{***}	0.02	-0.02^{*}	-0.02	-0.02^{*}	-0.01	-0.02	-0.02	0.34	-0.14	0.05	0.05
	(4.64)	(1.55)	(-1.70)	(-1.28)	(-1.91)	(-1.44)	(-0.37)	(-0.71)	(1.53)	(-0.99)	(0.84)	
Panel B: EUR Factors												
BTP Italia	1.65^{***}	-0.05^{**}	-0.06^{***}	-0.05^{**}	0.02	0.02	-0.01	0.03	0.05	-0.03	-0.03	0.11
	(3.36)	(-1.99)	(-2.97)	(-2.05)	(0.99)	(1.27)	(-0.08)	(0.36)	(0.11)	(-0.14)	(-0.27)	
CDS–Bond	5.14^{***}	0.00	0.02	0.04	-0.03	-0.03^{**}	0.47^{***}	-0.07^{*}	-0.53	0.15	-0.17	0.14
	(5.59)	(0.08)	(0.53)	(1.13)	(-1.38)	(-2.01)	(3.87)	(-1.70)	(-0.80)	(0.32)	(-1.05)	
Index Skew	2.26^{***}	-0.02	-0.04^{***}	-0.07^{***}	-0.03^{**}	0.01	0.06	-0.02	0.02	0.06	-0.07	0.09
	(4.64)	(-1.20)	(-2.60)	(-2.58)	(-2.27)	(1.16)	(1.00)	(-0.99)	(0.05)	(0.26)	(-0.90)	

Note: Monthly excess returns regressed on the ten Duarte et al. (2007) equity and bond risk factors. α is annualized (multiplied by 12). t-statistics in parentheses (Newey–West HAC). Mkt: market excess return (Fama–French). R_S: bank stock excess return. R_I, R_B: A/BAA-rated industrial and bank bond excess returns. R_2, R_5, R_{10}: maturity-sorted Treasury portfolio excess returns. Sample: BTP Italia 162 months, CDS–Bond 205 months, Index Skew 241 months. $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Table 28

Fung & Hsieh Factor Model Regressions

	α (%)	$\beta_{\text{S\&P}}$	$\beta_{\text{SC-LC}}$	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	β_{CredSpr}	$\bar{R}^2$
Panel A: US Factors									
BTP Italia	1.80***	−0.02*	−0.02**	0.12	-0.01	0.33	0.00	0.28	0.10
	(3.64)	(-1.96)	(-2.08)	(0.72)	(-0.03)	(1.05)	(0.01)	(1.18)	
CDS–Bond	5.28***	-0.02	0.01	0.09	-0.47	-0.09	−0.98***	−1.73***	0.18
	(6.68)	(-1.58)	(0.53)	(0.35)	(-1.49)	(-0.27)	(-3.98)	(-5.35)	
Index Skew	1.69***	0.02	−0.02**	0.10	0.17	0.13	-0.07	0.16	0.03
	(4.32)	(1.46)	(-2.09)	(0.48)	(0.89)	(0.48)	(-0.57)	(0.78)	
Panel B: EUR Factors									
BTP Italia	1.86***	−0.04***	−0.06**	0.34*	-0.08	0.34	0.28	-0.53	0.11
	(3.64)	(-2.99)	(-2.40)	(1.80)	(-0.39)	(0.92)	(1.44)	(-1.13)	
CDS–Bond	4.67***	-0.03	0.04	0.21	−0.71*	-0.59	−0.88**	-0.74	0.11
	(5.38)	(-1.60)	(1.22)	(0.88)	(-1.80)	(-1.64)	(-2.38)	(-1.32)	
Index Skew	3.48***	-0.00	−0.07**	0.12	-0.13	0.59*	-0.34	-0.12	0.05
	(6.05)	(-0.28)	(-2.02)	(0.53)	(-0.56)	(1.83)	(-1.37)	(-0.38)	

Note: Monthly excess returns regressed on the seven Fung & Hsieh (2004) hedge fund risk factors. α is annualized (multiplied by 12). t-statistics in parentheses (Newey–West HAC). S&P: S&P 500 excess return. SC–LC: Wilshire small-minus-large-cap equity spread. BdOpt, FXOpt, ComOpt: lookback-straddle returns on bond, currency, and commodity futures. 10Y: change in 10-year constant-maturity yield. CredSpr: change in Moody's Baa minus 10-year Treasury spread. Sample: BTP Italia 162 months, CDS–Bond 195 months, Index Skew 195 months. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.10$.

Table 29

Active Fixed Income Factor Model Regressions

	α (%)	β_{Term}	$\beta_{\text{Global Term}}$	$\beta_{\text{Global Agg}}$	$\beta_{\text{Infl. Linkers}}$	$\beta_{\text{Corp Credit}}$	$\beta_{\text{Emerg Debt}}$	$\beta_{\text{Emerg Curr}}$	$\beta_{\text{UST Vol}}$	$\bar{R}^2$
Panel A: US Factors										
BTP Italia	1.88***	0.15	-0.27	0.13	-0.02	−0.14**	0.09*	-0.02	0.00	0.11
	(3.65)	(0.98)	(-0.88)	(0.31)	(-0.82)	(-2.37)	(1.72)	(-0.91)	(0.15)	
CDS–Bond	4.61***	0.01	-0.20	0.42	-0.03	-0.00	0.07	0.02	0.01	0.10
	(5.38)	(0.08)	(-0.43)	(0.73)	(-0.44)	(-0.02)	(0.97)	(0.43)	(0.40)	
Index Skew	1.57***	-0.05	-0.10	0.21	-0.01	0.04	−0.07*	0.03	0.04**	0.10
	(4.36)	(-0.45)	(-0.43)	(0.68)	(-0.17)	(1.08)	(-1.82)	(1.49)	(2.00)	
Panel B: EUR Factors										
BTP Italia	1.96***	-0.07	-0.36	0.51*	−0.07**	-0.06	-0.03	0.03	0.00	0.01
	(3.28)	(-1.47)	(-1.43)	(1.87)	(-2.01)	(-1.23)	(-0.93)	(1.01)	(0.21)	
CDS–Bond	5.07***	-0.02	-0.05	0.25	-0.02	-0.08	0.15**	-0.02	-0.02	0.04
	(5.87)	(-0.17)	(-0.12)	(0.70)	(-0.24)	(-1.52)	(2.21)	(-0.31)	(-1.30)	
Index Skew	1.96***	0.03	-0.24	0.26	-0.02	0.01	-0.03	-0.02	−0.02*	0.01
	(4.73)	(0.32)	(-1.09)	(1.34)	(-0.50)	(0.81)	(-1.30)	(-1.16)	(-1.76)	

Note: Monthly excess returns regressed on the eight Active Fixed Income premia of Brooks et al. (2020). α is annualized (multiplied by 12). t-statistics in parentheses (Newey–West HAC). Term: Treasury excess returns (US or Eurozone). Global Term/Agg: Bloomberg Global Treasury/Aggregate (hedged). Infl. Linkers: Bloomberg Global Inflation-Linked. Corp Credit: HY and leveraged loans in excess of duration-matched Treasuries. Emerg Debt/Curr: EM bond and FX baskets. UST Vol: delta-hedged straddles on 10Y Treasury futures. Sample: BTP Italia 163 months, CDS–Bond 207 months, Index Skew 243 months. $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Table 30

Alpha Estimates Across Benchmark Factor Models

Panel A: Full-sample alpha (% p.a.)

Strategy	Duarte et al. (2007)			Brooks et al. (2020)			Fung & Hsieh (2004)		
	α	t	N	α	t	N	α	t	N
BTP Italia	$+1.65^{***}$	3.46	162	$+1.93^{***}$	3.19	163	$+1.86^{***}$	3.54	162
iTraxx Combined	$+2.26^{***}$	4.58	241	$+1.96^{***}$	4.50	243	$+3.48^{***}$	5.31	195
CDS–Bond Basis	$+5.14^{***}$	5.08	205	$+5.07^{***}$	5.28	207	$+4.67^{***}$	4.74	195

Panel B: Alpha by stress regime (% p.a.)

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL
BTP Italia	$+4.74^{**}$	$+1.65^{***}$	$+4.13^{**}$	$+1.42^{***}$	$+5.62^{***}$	$+1.46^{***}$
	(2.17)	(3.29)	(2.15)	(3.58)	(3.80)	(3.61)
iTraxx Combined	$+2.71^{***}$	$+1.98^{***}$	$+4.38^{***}$	$+1.38^{***}$	$+7.25^{***}$	$+2.61^{***}$
	(3.32)	(3.05)	(4.48)	(3.01)	(4.71)	(4.91)
CDS–Bond Basis	$+8.63^{***}$	$+3.77^{***}$	$+9.69^{***}$	$+3.63^{***}$	$+9.56^{***}$	$+2.95^{***}$
	(5.70)	(3.38)	(4.80)	(4.42)	(5.14)	(4.89)

Panel C: Ferson–Schadt (1996) conditional alpha (% p.a.)

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	α_1	t-stat	α_1	t-stat	α_1	t-stat
BTP Italia	$+1.09^{*}$	(1.66)	$+1.50^{**}$	(2.48)	$+1.39^{*}$	(1.95)
iTraxx Combined	$+1.06^{***}$	(3.59)	$+1.26^{***}$	(4.15)	$+2.81^{***}$	(4.63)
CDS–Bond Basis	$+2.87^{***}$	(2.97)	$+2.90^{***}$	(3.29)	$+3.31^{***}$	(3.12)

Note: Panel A reports annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors using EUR factors. Panel B reports alpha separately for HIGH and NORMAL regimes, where HIGH denotes months in which the 5-year iTraxx Europe Main spread exceeds 100 bps. Panel C reports the stress-loading coefficient α_1 from the Ferson and Schadt [1996] conditional model $r_t = \alpha_0 + \alpha_1 z_t + \beta' X_t + \varepsilon_t$, where z_t is the standardized iTraxx Main 5Y spread. $\alpha_1 > 0$: alpha increases during funding stress. t-statistics in parentheses. $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Table 31
PCA Spanning Regressions

	α (%)	β_{PC1}	β_{PC2}	β_{PC3}	β_{PC4}	β_{PC5}	β_{PC6}	β_{PC7}	β_{PC8}	$\bar{R}^2$	N
Panel A: Predictive ($PC_t \rightarrow R_{t+1}$)											
BTP Italia	1.68***	-0.012	-0.001	-0.026	-0.019	-0.012	-0.028	-0.001	-0.000	-0.005	156
	(4.05)	(-1.13)	(-0.07)	(-1.46)	(-0.61)	(-0.52)	(-1.46)	(-0.02)	(-0.02)		
CDS–Bond	5.43***	0.018	0.002	0.068*	0.059	−0.060**	0.011	-0.003	0.089***	0.050	196
	(5.91)	(1.23)	(0.09)	(1.69)	(1.17)	(-2.57)	(0.50)	(-0.08)	(3.03)		
Index Skew	2.07***	-0.011	-0.009	-0.002	0.013	0.023	0.015	0.026	0.042**	0.041	208
	(5.38)	(-1.20)	(-0.95)	(-0.13)	(0.78)	(0.89)	(0.62)	(1.47)	(2.11)		
Panel B: Contemporaneous ($PC_t \rightarrow R_t$)											
BTP Italia	1.61***	−0.025**	-0.010	-0.028	0.017	0.008	0.003	-0.013	0.010	0.054	157
	(3.48)	(-1.99)	(-0.97)	(-1.20)	(1.05)	(0.41)	(0.13)	(-0.50)	(0.60)		
CDS–Bond	5.87***	0.035***	0.010	0.067*	-0.016	-0.032	−0.079**	0.076**	0.051	0.120	197
	(5.74)	(3.07)	(0.52)	(1.95)	(-0.54)	(-0.80)	(-2.39)	(2.02)	(1.03)		
Index Skew	2.21***	0.001	-0.003	0.018	0.042	-0.007	0.012	−0.039**	0.018	0.049	209
	(5.01)	(0.16)	(-0.29)	(0.90)	(1.64)	(-0.54)	(0.65)	(-2.19)	(1.21)		

Note: Monthly strategy excess returns regressed on the first 8 principal components extracted from 24-month rolling windows. α is annualized (multiplied by 12). t-statistics in parentheses (Newey–West HAC). The first 8 PCs explain on average 80.8% of factor variance (min 75.7%, max 87.8%). Sample: BTP Italia 156 months, CDS–Bond 196 months, Index Skew 208 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 32
Top Factor Loadings by Principal Component

PC1		PC2		PC3	
Factor	Loading	Factor	Loading	Factor	Loading
EMERG_DEBT	+0.088	GLOBAL_AGG	+0.120	SS2Y	+0.062
CORP_CREDIT	+0.087	GLOBAL_TERM	+0.109	SS5Y	+0.053
ITRX_MAIN	-0.080	RI_EU	+0.109	PB_EU_CDS_5Y	+0.046
RS_EU	+0.079	Δ10Y_YIELD_EU	-0.104	ΔVIX	-0.042
DEF_US	+0.079	R2_EU	+0.103	PB_EU_CDS_1Y	+0.040
MKT_EU	+0.079	TERM_EU	+0.103	EP_SVIX_1M	+0.040

Note: Loadings are time-series averages of the eigenvector entries across all 24-month rolling windows. For each principal component, the 6 factors with the largest absolute loading are reported. Sign indicates the direction of the relationship.

Table 33

Adaptive Elastic Net: Post-Selection OLS on Stable Factors

	Coefficient	t-stat
Panel A: BTP Italia		
α (ann. %)	$+1.39^{***}$	3.34
SMB_EU	-0.044^{**}	-2.30
ILLIQ	$+0.638^{**}$	2.50
SS10Y	-0.021^{***}	-3.26
UMD_EU	$+0.030^{**}$	1.96
CDX_IG	$+0.009$	1.60
LIBOR_OIS	$+0.490^{*}$	1.90
$T = 157$, $k = 6$, $\bar{R}^2 = 0.2173$, DW = 2.0351		
Panel B: CDS–Bond Basis		
α (ann. %)	$+3.94^{***}$	3.66
ΔUF	-4.271^{***}	-2.61
RI_EU	$+0.160^{***}$	3.27
EMERG_FX	$+0.061^{**}$	1.99
PB_EU_CDS_1Y	$+0.011^{**}$	2.02
SS5Y	$+0.018^{**}$	2.33
CRED_SPR_US	-0.510^{**}	-2.20
EP_SVIX_1M	$+3.245^{***}$	2.82
PTFSFX	-0.450^{*}	-1.74
$T = 197$, $k = 8$, $\bar{R}^2 = 0.2670$, DW = 2.0233		
Panel C: iTraxx Combined		
α (ann. %)	$+0.47$	1.04
EP_SVIX_1M	$+3.306^{***}$	2.89
LIBOR_OIS	$+0.484^{**}$	2.21
ΔV2X	-0.017^{**}	-2.55
ΔUR	$+3.015^{**}$	2.35
ΔFAILS_PCT_TSY	$+0.402^{***}$	3.20
$T = 237$, $k = 5$, $\bar{R}^2 = 0.2359$, DW = 2.2489		

Note: Post-selection OLS on factors identified by bootstrap stability selection ($\hat{\Pi}_j \geq 0.80$). Factors are in original (un-standardized) units. α is annualized (% p.a.). Coefficients rounded to 3 decimals. t-statistics based on Newey–West HAC standard errors (6 lags). $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.10$. Factor definitions are provided in Table 5.

Table 34

Alpha by Stress Regime: PCA and AEN

	PCA		AEN	
Strategy	HIGH	NORMAL	HIGH	NORMAL
BTP Italia	$+5.37^{**}$	$+1.28^{***}$	+2.71	$+1.40^{***}$
	(2.24)	(4.21)	(1.43)	(3.64)
CDS–Bond Basis	$+9.13^{***}$	$+4.84^{***}$	$+8.17^{***}$	$+4.15^{***}$
	(3.77)	(4.43)	(4.62)	(4.41)
iTraxx Combined	$+3.38^{***}$	$+1.82^{***}$	$+1.51^{**}$	-0.94
	(4.36)	(4.22)	(2.03)	(-1.45)

Note: Annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors. PCA: spanning regression on 8 rolling principal components. AEN: post-selection OLS on bootstrap-stable factors. HIGH denotes months in which the 5-year iTraxx Europe Main spread exceeds 100 bps. t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 35
Factor Selection Method Comparison

Method	BTP Italia			CDS–Bond Basis			iTraxx Combined		
	k	α	$\bar{R}^2$	k	α	$\bar{R}^2$	k	α	$\bar{R}^2$
Lasso	7	$+1.39^{***}$ (3.39)	0.2203	11	$+3.91^{***}$ (3.44)	0.2594	13	$+0.67$ (1.58)	0.3016
Elastic Net	7	$+1.39^{***}$ (3.39)	0.2203	11	$+3.91^{***}$ (3.44)	0.2594	13	$+0.67$ (1.58)	0.3016
Adaptive Lasso	7	$+1.39^{***}$ (3.39)	0.2203	8	$+3.94^{***}$ (3.66)	0.2670	11	$+0.70$ (1.63)	0.3056
AEN	6	$+1.39^{***}$ (3.34)	0.2173	8	$+3.94^{***}$ (3.66)	0.2670	11	$+0.70$ (1.63)	0.3056
ALASSO-Chen	51	$+0.98$ (1.62)	0.4818	29	$+4.97^{***}$ (2.90)	0.3509	19	$+0.74^{*}$ (1.68)	0.3571

Note: α is annualized (% p.a.) from post-selection OLS. k denotes the number of selected factors. t-statistics (Newey–West HAC) in parentheses. AEN, LASSO, and Elastic Net use the same tuning criterion (HQC). Adaptive LASSO (Chen) uses AIC per Chen and Chen (2008). $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.10$.

Table 36

Unconditional and Regime-Dependent Correlations

Panel A: Strategy Returns

	Unconditional		By Regime		
Pair	Pearson	t-stat	LOW	MED	HIGH
CDS–Bond Basis vs BTP Italia	-0.183	-1.51	-0.077	-0.242^{**}	-0.625^{*}
CDS–Bond Basis vs iTraxx Combined	-0.019	-0.16	-0.117	-0.021	-0.255
BTP Italia vs iTraxx Combined	0.217^{***}	2.59	0.199	0.164	0.417

Panel B: Mispricing Changes (Δm)

	Unconditional		By Regime		
Pair	Pearson	t-stat	LOW	MED	HIGH
CDS–Bond Basis vs BTP Italia	-0.260^{***}	-2.89	-0.173	-0.291^{***}	-0.345
CDS–Bond Basis vs iTraxx Combined	0.182^{*}	1.65	0.099	0.098	0.380^{*}
BTP Italia vs iTraxx Combined	-0.152^{**}	-1.99	0.092	-0.169	-0.530

Note: Pearson correlations (unconditional) with HAC t-statistics (Newey–West). Regime columns: Pearson correlations in LOW, MEDIUM, and HIGH stress (iTraxx Main 5Y thresholds). Stars on unconditional: HAC inference. Stars on regime correlations: two-sided t-test with $n-2$ degrees of freedom. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table 37

Cross-Strategy Spanning Regressions (Δm)

Dependent	Regressor	β	t-stat	p	$\bar{R}^2$
CDS–Bond Basis	BTP Italia_L0	-0.232^{*}	-1.91	0.0560	0.0973
	BTP Italia_L1	-0.089	-0.87	0.3847	
	BTP Italia_L2	-0.069	-1.04	0.2980	
	iTraxx Combined_L0	$+1.270$	1.26	0.2060	
	iTraxx Combined_L1	$+1.344^{*}$	1.88	0.0597	
	iTraxx Combined_L2	$+1.081^{*}$	1.72	0.0858	
BTP Italia	iTraxx Combined_L0	-1.264^{**}	-2.52	0.0116	0.1328
	iTraxx Combined_L1	-1.558^{**}	-1.98	0.0472	
	iTraxx Combined_L2	$+0.407$	0.47	0.6406	
	CDS–Bond Basis_L0	-0.220^{***}	-2.64	0.0083	
	CDS–Bond Basis_L1	$+0.171^{**}$	2.21	0.0271	
	CDS–Bond Basis_L2	-0.037	-0.55	0.5789	
iTraxx Combined	BTP Italia_L0	-0.013	-1.18	0.2380	0.0545
	BTP Italia_L1	$+0.033$	1.60	0.1089	
	BTP Italia_L2	$+0.022^{**}$	2.04	0.0415	
	CDS–Bond Basis_L0	$+0.016$	1.08	0.2819	
	CDS–Bond Basis_L1	$+0.010$	0.86	0.3916	
	CDS–Bond Basis_L2	-0.009	-0.71	0.4774	

Note: $\Delta m_{i,t} = \alpha + \beta \Delta m_{j,t} + \varepsilon_t$. Newey–West HAC standard errors. $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Table 38

Stress Amplification: Interaction Regressions

Dependent	Variable	Coefficient	t-stat	p	$\bar{R}^2$
CDS–Bond Basis	Δm_j (BTP Italia)	-0.229	-1.60	0.110	0.0937
	$\Delta m_j \times z_t$	-0.061	-0.65	0.518	
	z_t (stress)	$+3.159^{**}$	2.49	0.013	
BTP Italia	Δm_j (CDS–Bond Basis)	-0.223^{**}	-2.32	0.021	0.0744
	$\Delta m_j \times z_t$	-0.156	-1.18	0.239	
	z_t (stress)	$+0.665$	0.34	0.736	
CDS–Bond Basis	Δm_j (iTraxx Combined)	$+0.746$	1.06	0.290	0.0585
	$\Delta m_j \times z_t$	$+0.186$	0.35	0.728	
	z_t (stress)	$+2.399^{***}$	2.72	0.006	
iTraxx Combined	Δm_j (CDS–Bond Basis)	$+0.017$	1.25	0.210	0.0460
	$\Delta m_j \times z_t$	$+0.023$	0.93	0.354	
	z_t (stress)	$+0.022$	0.09	0.926	
BTP Italia	Δm_j (iTraxx Combined)	-0.521	-1.01	0.314	0.0450
	$\Delta m_j \times z_t$	-1.265^{**}	-2.26	0.024	
	z_t (stress)	-0.071	-0.04	0.969	
iTraxx Combined	Δm_j (BTP Italia)	-0.013	-1.25	0.211	0.0482
	$\Delta m_j \times z_t$	-0.027^{***}	-3.11	0.002	
	z_t (stress)	$+0.184$	0.89	0.374	

Note: $\Delta m_{i,t} = \beta_0 \Delta m_{j,t} + \beta_1(\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t$, where z_t is the standardized iTraxx Main 5Y spread. $\beta_1 > 0$: interdependence increases under stress. Newey–West HAC standard errors. $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Table 39
AR(1) Persistence of Mispricing Levels

Strategy	ϕ	HAC t-stat	Half-life (months)	N
CDS–Bond Basis	0.8458***	20.28	4.1	148
BTP Italia	0.6829***	10.56	1.8	148
iTraxx Combined	0.5689***	5.99	1.2	148

Note: AR(1) regression of mispricing levels $m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}$, monthly frequency. Half-life $h_i^* = \ln(0.5)/\ln(\phi_i)$ measures the number of months for the mispricing level to halve following a shock. The CDS–Bond basis exhibits the highest persistence, consistent with its requirement of continuous repo financing of the cash bond [Boyarchenko et al., 2016]; the iTraxx skew, a pure derivative basis, exhibits the lowest. Newey–West HAC standard errors with six lags. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 40

Common Factor (PC1) and Funding Stress

Factor	Coefficient	t-stat	p-value
Panel A: Core Intermediary Proxies			
$T = 149$, $\bar{R}^2 = 0.231$			
HKM_IC	-4.292^{**}	-2.23	0.026
LIBOR_OIS	$+3.336^{***}$	3.07	0.002
PB_EU_CDS_5Y	-0.018^{**}	-1.97	0.049
ILLIQ	$+1.730^{***}$	2.79	0.005
Panel B: Extended Proxy Set			
$T = 149$, $\bar{R}^2 = 0.235$			
HKM_IC	-5.289^{***}	-2.85	0.004
LIBOR_OIS	$+3.247^{***}$	3.02	0.003
PB_EU_CDS_5Y	-0.011	-1.04	0.299
ILLIQ	$+2.007^{***}$	2.95	0.003
GC-REPO_T-BILL	-0.130	-0.35	0.727
BFCI_EU	$+0.527$	1.54	0.124
ΔFAILS_PCT_TSY	-5.490	-0.94	0.348
Panel C: Macroeconomic Placebo			
$T = 149$, $\bar{R}^2 = 0.112$			
ΔUM	$+0.234$	0.02	0.984
ΔUR	$+11.027$	1.30	0.194
5Y5Y_INFL	-0.579	-0.60	0.549
EPU_EU	$+0.003$	1.27	0.203

Note: OLS regression of the first principal component of the three Δm series on intermediary stress proxies and on a macro placebo set. Panel A: one proxy per Duffie channel (intermediary capital, funding cost, dealer health, market illiquidity). Panel B: all available funding and liquidity proxies. Panel C: macroeconomic placebo set (macro and real-activity uncertainty, long-term inflation expectations, economic policy uncertainty). Significant coefficients in Panels A and B indicate that the common factor driving correlated mispricings is linked to intermediary balance-sheet conditions [Duffie, 2010, Brunnermeier and Pedersen, 2009, He et al., 2017]. Non-significant coefficients in Panel C confirm that the common factor is not driven by macroeconomic fundamentals. Newey–West HAC standard errors. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.10$.

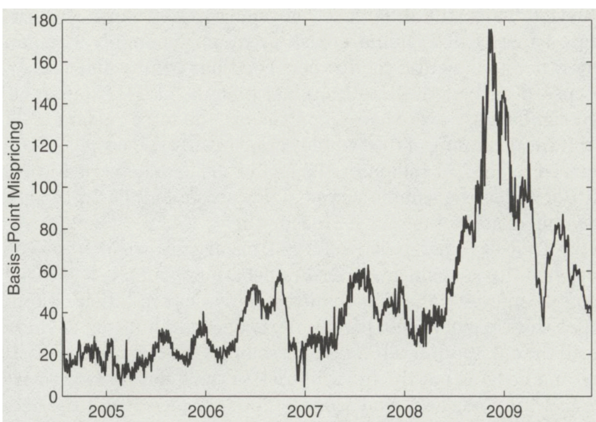

Figure 8
TIPS–Treasury basis, 2004–2009 (from Fleckenstein et al., 2014).

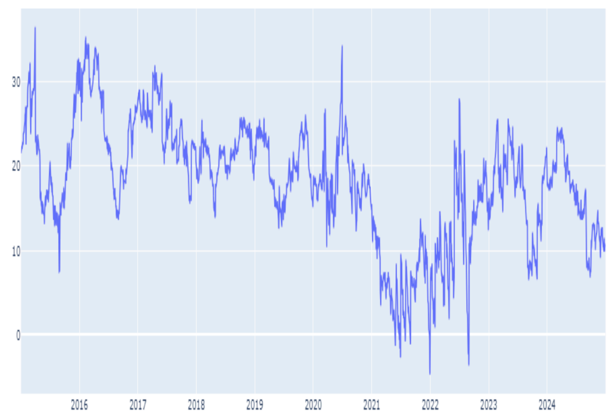

Figure 9
TIPS–Treasury basis, 2009–present. The mispricing has vanished.

Figure 10
Market-Level Basis Time Series for Fixed Income Arbitrage Strategies. For BTP Italia, the basis is the cross-sectional mean of all outstanding issues with at least six months to maturity. For iTraxx Combined, it is the mean of the four on-the-run sub-index skews. For CDS–Bond Basis, it is the mean of the $N = 20$ most negative single-name bases in the iTraxx Main universe on each day (only bonds with negative basis are included). Dashed lines indicate entry thresholds.

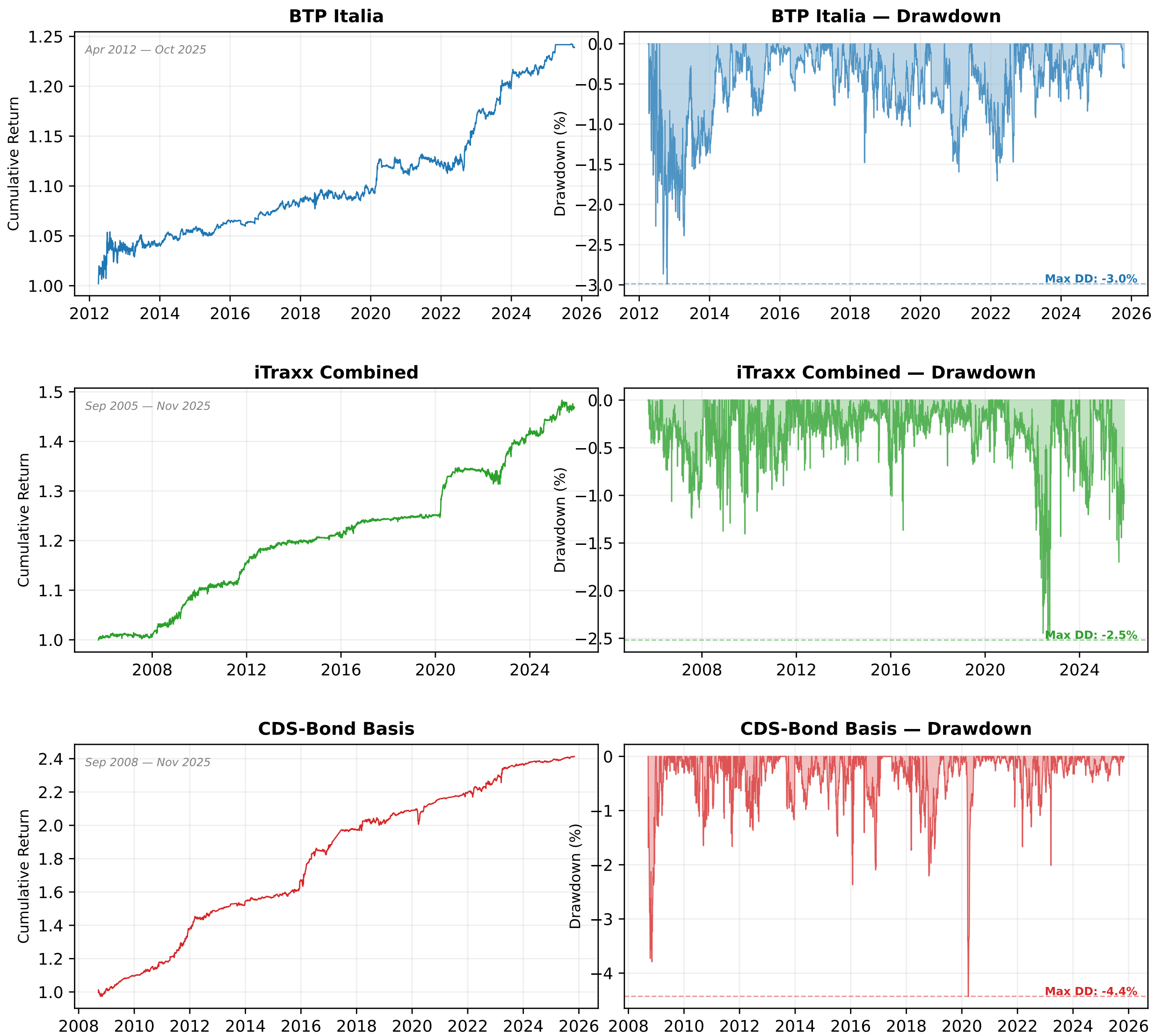

Figure 11

Cumulative Return Indices and Drawdowns of Fixed Income Arbitrage Strategies (Signal-Weighted). Each row shows one strategy: the upper sub-panel reports the cumulative return index from inception, and the lower sub-panel reports the running drawdown (peak-to-trough). Sample periods differ across strategies: BTP Italia from 2014, iTraxx Combined from 2008, CDS–Bond Basis from 2005. All series end in May 2025 and are net of transaction costs.

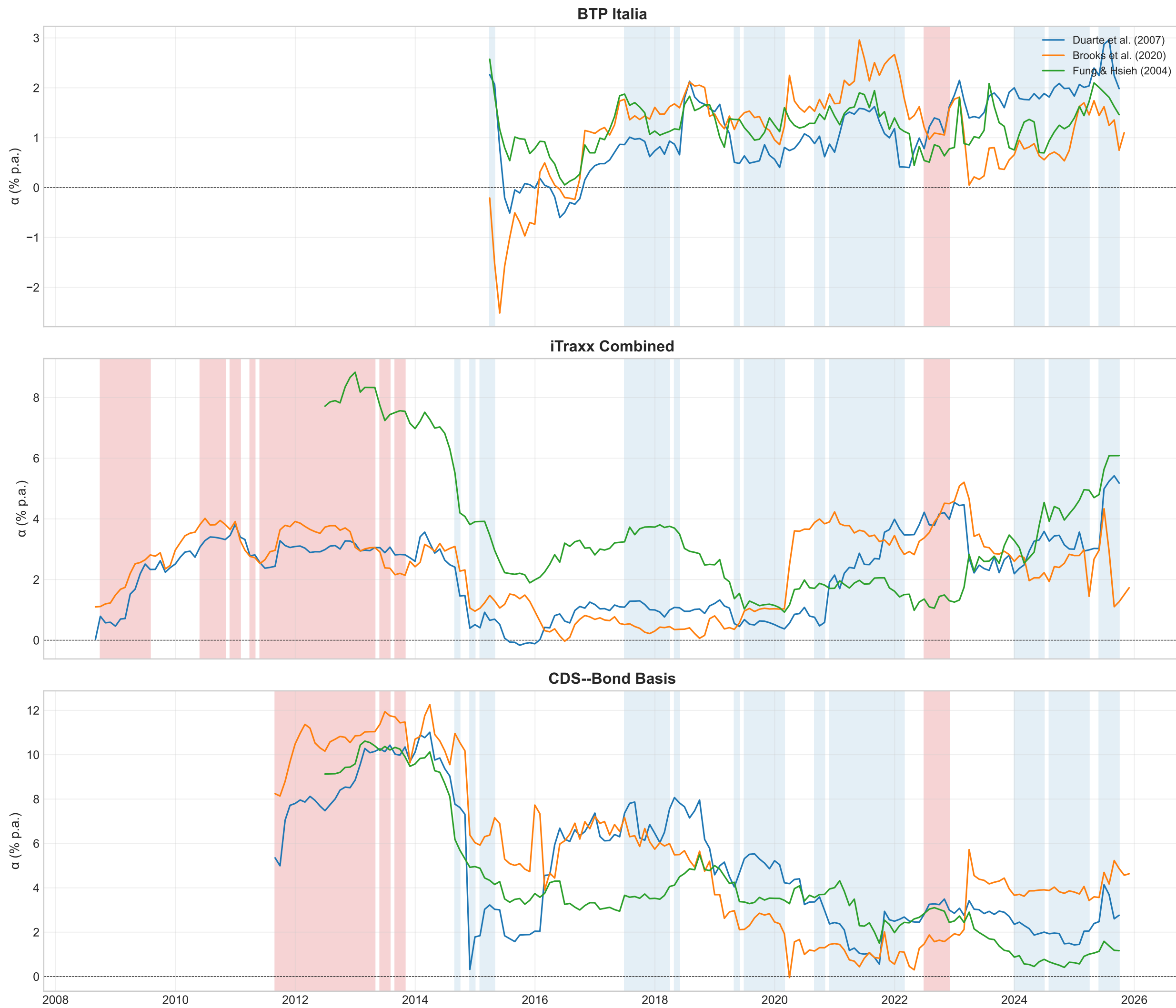

Figure 12
Rolling 36-Month Alpha Across Benchmark Factor Models. Each panel shows one strategy. Lines represent rolling alpha under Duarte et al. (2007, blue), Brooks et al. (2020, orange), and Fung & Hsieh (2004, green). Red shading: HIGH stress regime (iTraxx Main 5Y > 100 bps).

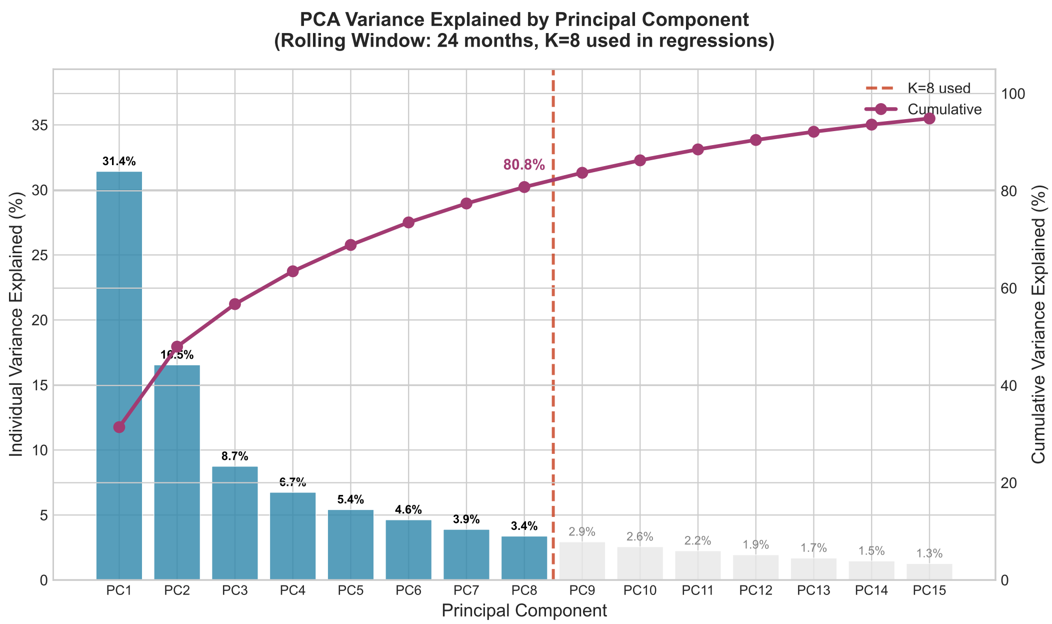

Figure 13
Scree Plot: Cumulative Variance Explained by Principal Components. Dashed line indicates 80% threshold. Baseline specification uses $K = 8$ components.

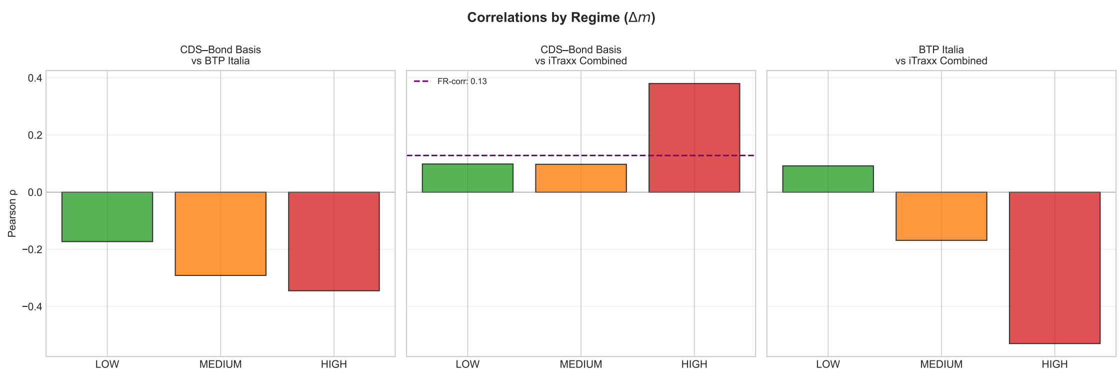

Figure 14

Pairwise Correlations by Stress Regime. Bars show Pearson correlations of Δm in LOW, MEDIUM, and HIGH stress regimes defined by iTraxx Main 5Y thresholds (60/120 bps).

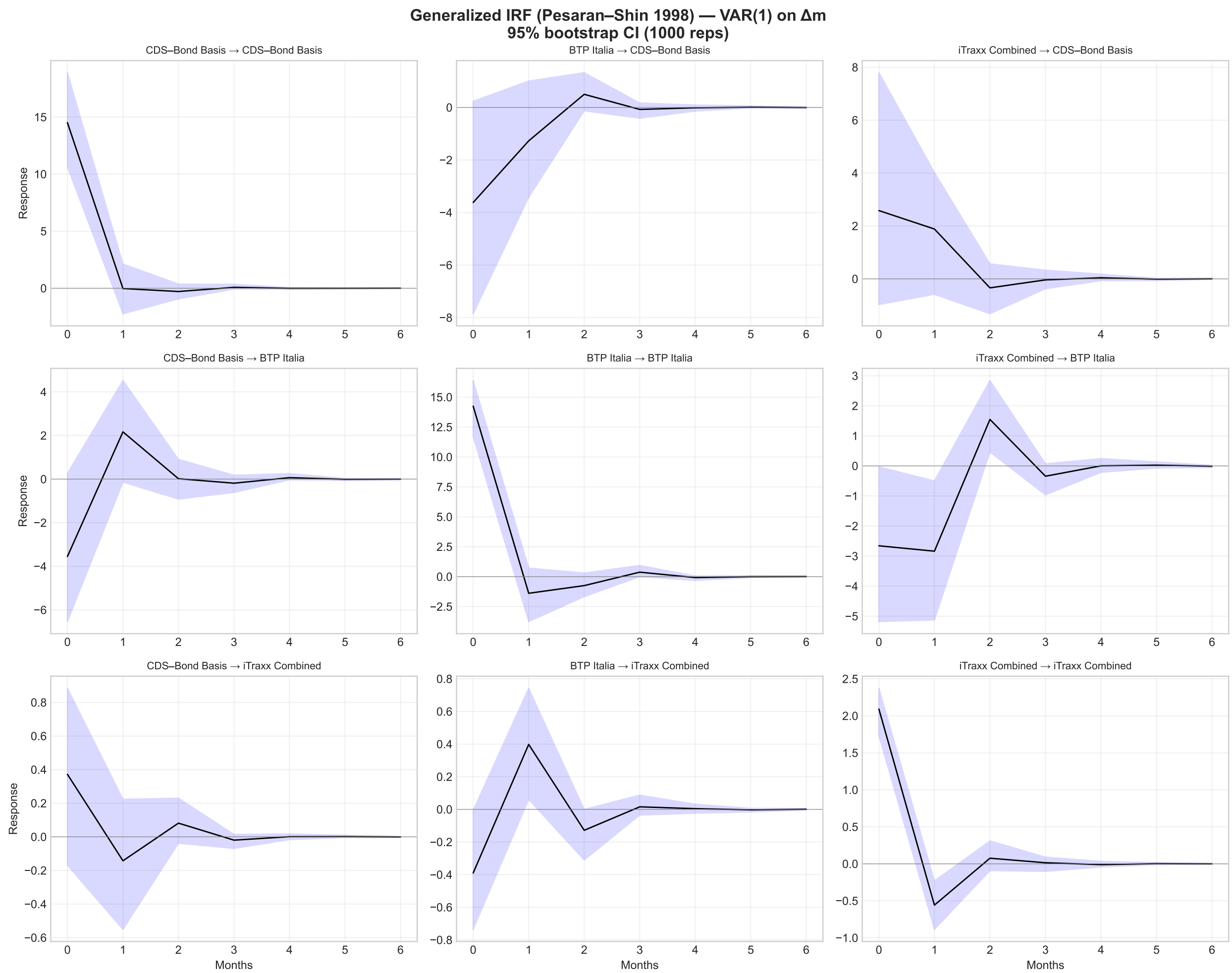

Figure 15
Generalized Impulse Response Functions. Responses of Δm to one-standard-deviation shocks. Shaded areas: 95% bootstrap confidence intervals (1000 replications).